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# Goal-Frontier Maximization as a Microfoundation for Welfare Economics

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## Abstract

We position Goal-Frontier Maximization as a measure-theoretic formalization of the capability welfare economics tradition of Sen, Nussbaum, Becker, and Stiglitz–Sen–Fitoussi: a tradition rich in qualitative content and in counting-based quantitative measurement (Alkire–Foster) but without a unified measure-theoretic substrate that admits cooperative excess and polarity discipline as first-class structure. The formalization yields two results of structurally distinct kinds: a *fungibility-collapse correspondence theorem* establishing GFM as a strict superset of standard utility theory at the linear / risk-neutral case, and a *Goodhart theorem* bounding each Lipschitz alignment property’s deviation from the operational truth (the realized exercise contribution  $T = \text{vol}_R^{[W]}$ ) by  $\text{Lip}(g) \cdot \varepsilon_{\text{gap}}^{\text{nonres}}$ , with property-specific Lipschitz analyses left to subsequent work. The proxy-truth gap admits four operational-intervention channels for tightening (observation density, attestation quality, individuation discipline, bundle decomposition), each formalizing a move the qualitative tradition has been negotiating informally. Admissible improvements to the economic model tighten certified alignment bounds in the direction of the operational truth, above the deployment-specific residual-class floor.

## 1 Introduction

Capability welfare economics has been a qualitatively rich tradition whose measure-theoretic apparatus has remained largely informal, even where its quantitative work is mature. Sen’s *Commodities and Capabilities* [Sen, 1985] introduces the capability set as the appropriate object for welfare analysis but does not endow it with a measure; the modern state of the qualitative tradition is surveyed in [Robeyns, 2017] and [Chiappero-Martinetti et al., 2020]. Nussbaum’s *Women and Human Development* [Nussbaum, 2000] adds substantive normative content but no structural axiomatization. Becker’s household-production model [Becker, 1965, 1981] specifies utility-yielding production functions that are descriptively flexible but predictively soft because the production functions are essentially unconstrained. The Stiglitz–Sen–Fitoussi commission [Stiglitz et al., 2009] catalogues at length what GDP-style measurement misses (non-market production, household labour, quality-of-life capabilities) but proposes only qualitative remedies (better surveys, satellite accounts, multi-dimensional indices) without a unified measure-theoretic framework on which the remedies compose. The Alkire–Foster multidimensional-poverty methodology [Alkire and Foster, 2011] supplies a quantitative apparatus on capability dimensions with deprivation thresholds and counting-based aggregation, but does not formalize cooperative excess across capability components or the polarity discipline that distinguishes cost-of-access from preference revelation.

The pattern is recognizable. Bentham’s felicific calculus, with its qualitative criteria of *intensity*, *duration*, *certainty*, *propinquity*, *fecundity*, *purity*, *extent*, was a qualitatively rich descriptive apparatus, and the expected-utility tradition (Bernoulli, Marshall, Edgeworth, von Neumann–Morgenstern) formalized the qualitative claims into a measure-theoretic apparatus admitting axiomatization, falsification, and quantitative comparison. We position the Goal-Frontier Maximization

(GFM) sequence [Lasser, 2026b,g,c,h,a,f,i,d,e] as the analogous formalization for capability welfare economics. The poset structure formalizes Sen’s capability set; the volume measures  $\text{vol}_P$  and  $\text{vol}_R^{[W]}$  formalize the capability/functioning distinction; the residual class ResS formalizes the shadow-economy / non-market-activity gap; the five-cell  $\delta$ -decomposition formalizes the SSF beyond-GDP catalogue. The sequence has proved its central alignment results (the anti-monopolar property [Lasser, 2026c, Horizon Aware], the phase boundary [Lasser, 2026f, Phase Redundancy], and controlled-relaxation convergence [Lasser, 2026i, Controlled Relaxation]) under the standing assumption that  $\text{vol}_P$  is the operational target. The B-to-C gap, named in [Lasser, 2026b, Section 7] and addressed at length in [Lasser, 2026d, Revealed Sacrifice] and [Lasser, 2026e, Need Sufficiency], is the divergence between  $\text{vol}_P$  (capability *possession*) and  $\text{vol}_R$  (capability *exercise*). Every result in the sequence inherits this gap as a structural dependency: if the gap is large, the guarantees the framework establishes on  $\text{vol}_P$  may fail to translate to guarantees on what the collective actually does.

This paper makes the dependency formal, reading the B-to-C gap as the central object that the formalization makes computable and bounding the cost of the gap to alignment guarantees in the sequence. The formalization yields two formal contributions of structurally distinct kinds: a *containment proof* establishing GFM as a strict superset of standard utility theory at the linear / risk-neutral case, and an *alignment bound* formalizing the cost of proxy-instrumental optimization on GFM’s volume measure.

1. A *fungibility-collapse correspondence theorem* (Section 2) showing that GFM strictly contains standard microeconomic utility theory in its linear-utility / risk-neutral form: utility theory emerges as the special case of GFM where the capability poset reduces to a single purchasing-power dimension, cooperative capabilities vanish, and individuation becomes irrelevant. The result is positioning, not an alignment guarantee: it establishes containment (utility theory  $\subset$  GFM at  $P_{\text{std}}$ ) plus extension (GFM has five structural features beyond  $P_{\text{std}}$  that utility theory lacks). Concavity, risk-aversion, and heterogeneous utility require additional calibrations beyond the bare collapse, each a small step beyond the correspondence rather than a new theorem.
2. A *Goodhart theorem for proxy-instrumental alignment* (Section 4). When the framework optimizes a proxy  $P = \text{vol}_P$  for the operational truth  $T = \text{vol}_R^{[W]}$ , and  $P$  is related to  $T$  only instrumentally, each alignment property  $g(\cdot)$  in the sequence, read as a Lipschitz function of  $P$ , deviates from its value under  $T$  by at most  $\text{Lip}(g) \cdot \varepsilon_{\text{gap}}^{\text{nonres}}$ , where  $\varepsilon_{\text{gap}}^{\text{nonres}}$  is the within-channel proxy-truth disagreement gap. This is the paper’s load-bearing alignment result. The property-specific Lipschitz analyses (anti-monopolar threshold, phase-boundary location, convergence rate, damage bound, aggregate-lower-bound slack) are perturbative roadmap items for subsequent work; this paper establishes the conditional transfer and identifies which results live in the roadmap.

Plus a *strong-form treatment* (Section 5) of what “better economic models” means structurally. “Better economics” is not one knob but four operational-intervention channels (observation density, attestation quality, individuation discipline, bundle decomposition), each formalizing a move the qualitative welfare-economics tradition has been negotiating informally, and each composing through the Goodhart machinery into a specific alignment-tightening consequence. The capability-economics lineage of Sen, Nussbaum, Becker, and SSF maps onto these channels as historical confirmation: each qualitative tradition is a precursor whose negotiation GFM formalizes. Section 5.7 also situates GFM relative to the broader formal-welfare-economics landscape, including Alkire–Foster’s quantitative comparator, Bernheim–Rangel’s behavioral welfare economics, Basu–López-Calva’s prior formalization survey, and the Pattanaik–Xu / Sugden / Adler axiomatic-aggregation neighbors. Better economic models are tighter alignment guarantees, above the deployment-specific residual-class floor where capabilities exercised without material or time-displacement footprint structurally cannot be witnessed.

## 1.1 What was missing from the picture

[Lasser, 2026d, Revealed Sacrifice] establishes that voluntary trade events disclose  $\text{vol}_R$ -content through a privacy-respecting channel, and that the constructed lower bound  $\text{vol}_R^{\text{lower}}$  on the realized exercise volume is non-decreasing in admissibly accumulated observations. [Lasser, 2026e,

Need Sufficiency] adds the diagnostic architecture: the five-cell decomposition of the gap, the wireheading-consistent concentration signal, the residual-class characterization, and the polarity discipline that distinguishes need-side from want-side trades. What was missing was the formal link between *quality of the economic model* and *tightness of the alignment guarantees*. If the framework optimizes  $\text{vol}_P$  but the truth is  $T$ , and if  $\text{vol}_P$  is only instrumentally related to  $T$ , then any alignment property proved on  $\text{vol}_P$  need not transfer to  $T$ .

The Goodhart theorem of Section 4 makes this transfer-or-failure formal: it gives a Lipschitz-style continuity statement showing that alignment properties under the truth deviate from those under the proxy in proportion to the proxy-to-truth gap. The B-to-C gap is therefore not a binary precondition but a continuous gradient: what looked like a single “does the gap close?” question is, on inspection, a *rate-of-closure-vs-residual-floor* question that the machinery of [Lasser, 2026d, Revealed Sacrifice] and [Lasser, 2026e, Need Sufficiency] is built to answer. The four channels of Section 5 are how the gap closes; the floor is what closing saturates against.

## 1.2 The operational truth $T$ , and the residual-class floor

A clarification before the central machinery. Under [Lasser, 2026d, Revealed Sacrifice]’s assumption S0 (valuation bounded by exercise: an agent’s valuation  $U_i(Y)$  does not exceed the window-active realized exercise contribution  $\text{vol}_R^{[W]}(Y)$ ), trade events provide a one-sided constraint relating proxy disagreement to realized-exercise truth. We adopt  $T(c) = \text{vol}_R^{[W]}(c)$  as the *operational truth* measure throughout this paper.  $T$  is what [Lasser, 2026d, Theorem 2] bounds from below via  $\text{vol}_R^{\text{lower}}$ , and it is what alignment properties operate on in practice. The metaphysical question (“what is the agent’s true inner preference?”) is unobservable in principle, and the framework does not pretend to access it. S0 is the operational bridge.

The proxy-to-truth gap has two structurally distinct components. The *within-channel disagreement*  $\varepsilon_{\text{gap}}^{\text{nonres}}$  is the worst-case relative deviation of the proxy from the truth on capabilities the revealed-sacrifice channel can in principle reach; this component is drivable downward by observation infrastructure. The *residual-class floor*  $\varepsilon_{\text{floor}}^{\text{res}}$  is the share of operational truth that lives in capabilities structurally outside the channel’s reach (purely-private exercise with no material *and* no time-displacement footprint); this component cannot be reduced by within-channel observation alone. The total alignment slack is

$$\varepsilon_{\text{gap}} = \varepsilon_{\text{gap}}^{\text{nonres}} + \lambda \cdot \varepsilon_{\text{floor}}^{\text{res}},$$

with  $\lambda \in [0, 1]$  weighting the floor’s contribution.

The split is structurally important and ties the alignment programme directly to a long-standing welfare-economics problem. Standard economics has the same residual-class limit [Stiglitz et al., 2009]: GDP misses non-market production, household labour, volunteer activity, gift-economy capabilities. Imputation methods (imputed rent, time-use surveys, satellite accounts) are the existing partial answer. GFM does not remove the limit; it makes the limit formally identifiable as the structural residual class ResS of [Lasser, 2026e, Definition 12] with a reserved cell ( $\delta_{\text{residual}}$ ) in the gap decomposition. The fungibility-collapse correspondence of Section 2 simplifies the poset structure but does *not* extend the channel’s reach. The shadow-economy gap and GFM’s residual class are the same structural object viewed from different sides.

## 1.3 Scope: what this paper does not do

- **It does not prove the full perturbation analysis.** This paper establishes the *conditional* Lipschitz-transfer result (Theorem 2) and identifies the alignment properties (anti-monopolar threshold, phase-boundary location, convergence rate, damage bound, aggregate-lower-bound slack) as the *perturbative roadmap*: candidates for a subsequent perturbation analysis. Each property’s Lipschitz constant requires its own analysis, which is outside this paper’s scope. We give the qualitative direction, indicate where each constant lands, and provide concrete worked examples of the tightening behaviour.
- **It does not characterize the rate of  $\varepsilon_{\text{gap}}^{\text{nonres}}$  convergence.** [Lasser, 2026d, Propositions 2 and 3] prove  $\beta^{\text{lower}}$  is non-decreasing under admissible accumulation; they do not bound *how fast*  $\beta^{\text{lower}}$  converges. The rate depends on S0 attestation quality, Part-A vs Part-B aggregation choice, observation density, and the residual-class floor itself. Bounding the

rate is [Lasser, 2026e, Need Sufficiency]’s Open Question 6 and is the territory of subsequent perturbative analysis; Corollary 1 of the present paper is a direction (monotonicity), not a rate.

- **It does not resolve the metaphysical truth question.** The Goodhart machinery operates on the operational  $T = \text{vol}_R^{[W]}$  under S0, not on a metaphysical model of inner preference. Whether the agent’s “true inner preference” (in some philosophical sense) coincides with  $\text{vol}_R^{[W]}$  is outside the formal apparatus.
- **It does not provide a normative capability list.** GFM provides the structural scaffold on which normative choices can be loaded; the choice of *which* capabilities count as central is outside the formal apparatus. This is consistent with Sen’s stance in the original capability approach [Sen, 1985]; Nussbaum’s substantive list [Nussbaum, 2000, 2011] is one normative loading, not the only one.
- **It does not unify the alignment and welfare-economics programmes.** The two programmes share structural content (poset, capability measure, residual-class limit), but their goals differ (alignment is about constraining optimization; welfare economics is about evaluating distributions). The formalization claim is that both programmes have the same structural substrate, and progress on either drives down the same  $\varepsilon_{\text{gap}}^{\text{nonres}}$  that the other depends on; this is structural co-progress, not unified goals.
- **It does not directly address Arrow’s impossibility [Arrow, 1951].** GFM measures capability-volume rather than ranking outcomes, so the independence-of-irrelevant-alternatives axiom that drives Arrow’s impossibility does not apply to a volume measure on a poset. A careful argument identifying which Arrowian axioms GFM satisfies, which it appears to sidestep at the level of Arrow’s original formulation, and where this places the framework relative to the social-choice literature [Sen, 1970] is in Section 5.7, with the question of whether an analogous impossibility applies to the relaxed setting deferred to Open Question 8.4.

## 1.4 Summary of results

1. **Fungibility-collapse correspondence (containment result)** (Section 2). Theorem 1: an actor maximizing  $\text{vol}_P$  on the standard-economics poset  $P_{\text{std}}$  is observationally equivalent to a linear-utility / risk-neutral expected-utility maximizer with the same discount factor. Together with the five-features catalog (non-fungibility, cooperative capabilities, anti-monopolar reasoning, individuation without interpersonal comparison, and formal recognition of the residual class), this establishes that GFM strictly contains standard utility theory at the linear / risk-neutral case. Concavity, risk-aversion, and heterogeneous utility recover through additional calibrations.
2. **Structural-prediction class** (Section 3). Five predictions GFM makes that utility theory either does not, or can match only by adding ad hoc unfalsifiable structure: bundle-completion magnitude, cooperative premium, saturation transition, population-scale anti-monopolar (via the wireheading-consistent HHI [Lasser, 2026e, Proposition 6]), and mixed-polarity bundle-for-bundle traces (via the bundle-decomposition extension flagged in [Lasser, 2026d, Open Question 2]). These are what becomes falsifiable when capability welfare economics is quantified into a measure-theoretic apparatus.
3. **Goodhart theorem for proxy-instrumental alignment** (Section 4). Theorem 2 establishes the conditional Lipschitz-transfer  $|g(T) - g(P)| \leq \text{Lip}(g) \cdot \varepsilon_{\text{gap}}^{\text{nonres}}$ . This is the paper’s novel formal contribution. A separate qualitative claim (Conjecture 1) states that more capable optimizers exploit larger sets of proxy-truth divergence regions, widening the observed  $\varepsilon_{\text{gap}}^{\text{nonres}}$  at equilibrium; we do not prove this conjecture, but the wireheading-consistent HHI is a deployment-readable “concentration consistent with” empirical witness for the conjectured mechanism.
4. **Strong-form direction: better economics tightens alignment** (Section 5). Corollary 1 states the direction (monotonicity); Proposition 1, conditional on per-capability admissibility (Assumption 1), composes four operational-intervention channels into an aggregate alignment-slack reduction. Channel 3 contributes through the  $\lambda$ -weighted net  $\Delta_3^\lambda = \lambda \cdot \Delta_3^{\text{floor}} - \Delta_3^{\text{new}}$ , capturing the floor reduction from re-individuation against the new

within-channel ceiling contribution from the newly-individuated capability. The lineage subsection (Section 5.7) maps each precursor onto the channels: the qualitative-tradition figures (Sen, Nussbaum, Becker, SSF, Arrow) supply the moves the channels formalize, and the already-quantitative comparator (Alkire–Foster multidimensional poverty measurement, [Alkire and Foster, 2011]) supplies the closest existing apparatus that GFM extends with cooperative excess and polarity discipline.

## 1.5 Design principle

This paper inherits the structural discipline of the preceding sequence papers and applies it one level up. [Lasser, 2026f, Phase Redundancy] and [Lasser, 2026i, Controlled Relaxation] gain precision through frequency rather than depth (more tests, not more-intrusive tests). [Lasser, 2026d, Revealed Sacrifice] gains exercise-volume precision through frequency rather than depth (more trade observations, not more-intrusive trader surveillance). The present paper inherits a meta-version of the same discipline: *the quality of the economic model controls the tightness of the alignment guarantees, and the quality of the economic model is itself a question of channel reach and observation density*, not of more-intrusive measurement. The proxy-to-truth gap is closed by observation infrastructure operating within the framework’s existing privacy commitments, not by relaxing those commitments.

## 1.6 Dependencies on prior papers

A summary of the prior-paper results this paper relies on is collected in Appendix A. The companion papers [Lasser, 2026d, Revealed Sacrifice] and [Lasser, 2026e, Need Sufficiency] and the earlier sequence papers [Lasser, 2026b,g,c,h,a,f,i] are presently distributed as preprints. The propositions and definitions imported here are pinned by their current formulations; if the upstream preprints revise their numbering or content, downstream references will need refreshing correspondingly. Stable archival identifiers will be added as the sequence reaches archival status.

# 2 The Fungibility-Collapse Correspondence

The introduction posed GFM as a measure-theoretic formalization of capability welfare economics. This section establishes the *containment* half of that positioning: the formal proof that standard utility theory sits inside GFM as a strict subset at the linear / risk-neutral case. Standard utility theory, in its linear / risk-neutral form, emerges as the special case of GFM under what we call *fungibility collapse*: the structural reduction in which the capability poset degenerates to a single purchasing-power dimension, cooperative capabilities vanish, and individuation becomes irrelevant. The correspondence is a containment result rather than an alignment guarantee: it tells the reader where standard economics sits inside the apparatus, what additional structure GFM brings beyond it, and where the two frameworks coincide versus diverge. The load-bearing alignment result, the Goodhart theorem of Section 4, operates against this established containment as backdrop.

The reduction is exact only in the linear / risk-neutral case. Recovering broader content of standard utility theory (concave  $u$ , risk-aversion, heterogeneous agents) requires *additional* calibrations beyond the bare collapse, each a small step rather than a new theorem. We make these scope conditions explicit in Section 2.3.

A structural caveat carries through the entire section: fungibility collapse simplifies the *poset structure* (cooperative capabilities, individuation, non-fungibility) but does *not* extend the observation channel’s reach. The residual class of [Lasser, 2026e, Need Sufficiency], Definition 12, remains non-empty under collapse: standard economics has the same shadow-economy / non-market-activity gap that [Stiglitz et al., 2009] documents, and the formal residual class is the GFM-side counterpart to that gap. Section 2.4.5 makes this caveat precise.

## 2.1 The standard-economics poset $P_{\text{std}}$

**Definition 1** (Standard-economics poset). *The standard-economics poset  $P_{\text{std}}$  is the totally-ordered single-dimension capability space*

$$P_{\text{std}} = \{d \in \mathbb{R}_{\geq 0} : d \text{ is a unit of purchasing power}\},$$

with the standard real-line ordering  $d \leq d'$  iff  $d \leq_{\mathbb{R}} d'$ , weight function  $w(d) = d$  (money is its own measure), and the following three structural reductions:

- (P1) **No cooperative capabilities.** The cooperative- capability term in axiom M6 ([Lasser, 2026g, Proposition 1]) vanishes: superadditive combinations of capabilities reduce to additive ones. The only interaction between capability units is exchange at market prices, which is captured at the macro level by the additivity of the weight function; no  $w(\{d_1, d_2\}) > w(d_1) + w(d_2)$  structure is admitted.
- (P2) **No individuation.** Agents are anonymous / representative. Per-agent benchmark calibrations  $w_i(\cdot)$  reduce to the common  $w(\cdot)$ , and questions of interpersonal comparison (which the GFM individuation discipline of M5 keeps separate) are absent at  $P_{\text{std}}$ 's level of structure.
- (P3) **Total ordering.** Any two capability units  $d, d' \in P_{\text{std}}$  are comparable by the real-line ordering, so the poset's lattice structure degenerates to a chain. No incomparable capabilities exist; no diversity dimension is preserved.

**Remark 1** (Fungibility collapse as the conditions producing  $P_{\text{std}}$ ). We use “fungibility collapse” throughout to denote the structural reduction conditions (P1)–(P3) under which the GFM apparatus operates on  $P_{\text{std}}$ . The collapse is not an operation performed on a specific GFM poset  $P$ ; it is a statement about the regime in which the non-fungibility, cooperative, and individuation features of  $P$  are operationally insignificant and the capability space is approximately  $P_{\text{std}}$ . At macro scales with liquid markets, approximately fungible goods, and representative-agent reasoning, this regime is a reasonable approximation; at micro scales, in non-market sectors, or in domains with strong cooperative dynamics, the regime fails and the additional GFM structure becomes load-bearing (Section 2.4).

## 2.2 The correspondence theorem

**Theorem 1** (Fungibility-Collapse Reduction, linear-utility special case). Let  $\{S_t\}_{t \geq 0}$  be an adapted, measurable capability-holding process on a filtered probability space  $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}, \mathbb{P})$  describing the framework's dynamics, and assume the discounted volume sum is integrable (an absolute summability condition that holds under  $\gamma < 1$  and almost-sure boundedness of the period volumes). Then an actor maximizing  $\text{vol}_P$  on  $P_{\text{std}}$  with discount factor  $\gamma \in (0, 1)$  is observationally equivalent, in the sense of producing the same choice sequence under the same observable constraints, to an expected-utility maximizer with utility  $u(x) = x$  (linear utility over wealth, equivalently risk-neutral) and discount factor  $\gamma$ .

*Proof.* We verify the equivalence by computing the actor's objective under each interpretation and showing they coincide.

$\text{vol}_P$  on  $P_{\text{std}}$ . By Definition 1, the cooperative-capability term in axiom M6 vanishes. Axiom M4 (additivity over disjoint subsets of capabilities) of [Lasser, 2026g, Proposition 1] then gives, for any subset  $S \subseteq P_{\text{std}}$  measurable under the volume measure,

$$\text{vol}_P(S) = \int_S w(d) d\mu(d) = \int_S d d\mu(d),$$

where  $\mu$  is the underlying counting / Lebesgue measure on  $P_{\text{std}}$ . In the discrete reading (capability units are atomic),  $\text{vol}_P(S)$  is the sum of unit values, i.e., the total purchasing power held in  $S$ . In the continuous reading,  $\text{vol}_P(S)$  is the real-valued integral over  $S$ . In either case,  $\text{vol}_P$  on  $P_{\text{std}}$  reduces to total wealth.

**Discounted maximization.** The actor's objective, summed over a horizon  $\{0, 1, 2, \dots\}$  with discount factor  $\gamma$ , is the expected discounted volume

$$\text{Obj}_{\text{vol}_P} = \mathbb{E} \left[ \sum_{t=0}^{\infty} \gamma^t \text{vol}_P(S_t) \right],$$

where  $\{S_t\}_{t \geq 0}$  is the adapted, measurable capability-holding process on a filtered probability space  $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}, \mathbb{P})$  describing the framework's dynamics (deterministic or stochastic). We assume the

realized objective is integrable (an absolute summability condition that holds under  $\gamma < 1$  and almost-sure boundedness of the period volumes; the assumption is standard for discounted infinite-horizon objectives). Under fungibility collapse the volume on  $P_{\text{std}}$  reduces to the wealth integral and the objective rewrites as

$$\text{Obj}_{\text{vol}_P} = \mathbb{E} \left[ \sum_{t=0}^{\infty} \gamma^t \int_{S_t} d \, d\mu(d) \right] = \sum_{t=0}^{\infty} \gamma^t \mathbb{E}[W_t],$$

where  $W_t$  is total wealth held at time  $t$  and the exchange of  $\mathbb{E}$  and  $\sum$  uses Fubini/Tonelli under the integrability assumption. Under deterministic dynamics the expectation is the identity and  $\text{Obj}_{\text{vol}_P} = \sum \gamma^t W_t$ .

**Expected-utility comparator.** A linear-utility / risk-neutral expected-utility maximizer with discount factor  $\gamma$  has objective

$$\text{Obj}_{\text{EU}} = \sum_{t=0}^{\infty} \gamma^t \mathbb{E}[u(W_t)] = \sum_{t=0}^{\infty} \gamma^t \mathbb{E}[W_t].$$

The two objectives coincide:  $\text{Obj}_{\text{vol}_P} = \text{Obj}_{\text{EU}}$ . Under deterministic dynamics this is the per-realization equality  $\sum \gamma^t W_t$ . Under stochastic dynamics, both objectives reduce to  $\sum \gamma^t \mathbb{E}[W_t]$ , so the agent’s optimal choice sequence is determined by the same sufficient statistic in either interpretation: total expected discounted wealth.

**Anti-monopolar property degenerates.** The anti-monopolar property of [Lasser, 2026c, Proposition 6] penalizes full domination by a single substrate, and is non-trivial when multiple capability dimensions exist. On  $P_{\text{std}}$ , only one dimension is present; “domination” has no other dimension to contrast against, and the anti-monopolar correction vanishes identically.

**Conclusion.** The actor’s choice sequence under either interpretation is determined by the same sufficient statistic (total expected discounted wealth), so the two interpretations are observationally equivalent. ■

**Remark 2** (“Observationally equivalent” is not “the same theory”). *Theorem 1 says the actor’s behavioural predictions coincide on  $P_{\text{std}}$ , not that GFM and standard utility theory are the same theory. The frameworks differ in what they say about the agent: GFM’s volume measure has axiomatic structure (M1–M6) that survives off  $P_{\text{std}}$ ; standard utility theory’s  $u(\cdot)$  does not. The frameworks coincide on  $P_{\text{std}}$  at the level of what choices to predict; they diverge on which apparatus carries the prediction’s structure as soon as one steps off  $P_{\text{std}}$ . We belabour the distinction because the positioning claim of this paper rests on it: GFM is the formal descendant of standard utility theory at  $P_{\text{std}}$ ; it is also a strictly stronger structure off  $P_{\text{std}}$ , in the sense that the poset, cooperative, individuation, and residual-class machinery have content that utility theory does not.*

### 2.3 Scope: what additional structure recovers more of utility theory

Theorem 1 covers the linear / risk-neutral case. Standard microeconomic utility theory is broader: concave  $u$ , risk-aversion, and heterogeneous agents are routine. The reductions that recover each in the GFM apparatus are small steps beyond the bare collapse, each requiring an additional calibration on top of  $P_{\text{std}}$ .

**Remark 3** (Concave utility). *A concave  $u : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}$ , capturing diminishing-returns-on-wealth preferences, corresponds in the GFM apparatus to a re-weighting  $w'(d) = u(d)$  on the single-dimensional axis. This is a recalibration of the benchmark function (moving from the identity weight  $w(d) = d$  to a concave  $w' = u$ ) rather than a new poset structure.  $\text{vol}_P$  evaluated with  $w'$  on  $P_{\text{std}}$  is then  $\int_S u(d) \, d\mu(d)$ , equal to the expected-utility objective under  $u$ . GFM’s individuation discipline (M5) requires that this re-weighting be applied consistently across agents at the positioning level; a per-agent re-weighting goes to the heterogeneity discussion below. The recovery is structural: any calibrated GFM apparatus on  $P_{\text{std}}$  with weight  $w' = u$  matches the corresponding expected-utility maximizer.*

**Remark 4** (Risk-aversion). *Risk-aversion in the standard sense (preferring a deterministic outcome to a lottery with the same expected value) corresponds in the GFM apparatus to additional structure*

on the bundle’s risk-distribution beyond the deterministic dynamics assumed in the proof of Theorem 1. The controlled-relaxation machinery of [Lasser, 2026i, Controlled Relaxation] treats risk explicitly via the risk-claim landscape and the convergence dynamics that follow from it. Composing the calibrated  $w' = u$  of Remark 3 with that machinery, and additionally imposing the standard expected-utility independence axiom on lotteries, reproduces risk-averse expected-utility maximization with concave  $u$ . No new theorem is required at this paper’s level of analysis; the composition is mechanical given the calibration choices.

**Remark 5** (Heterogeneous utility across agents). *Heterogeneous  $u_i(\cdot)$  across agents corresponds in the GFM apparatus to per-agent benchmark calibrations  $w_i(d)$  on the same underlying  $P_{\text{std}}$ . GFM’s individuation-without-interpersonal-comparison stance ([Lasser, 2026g, Proposition 1], axiom M5) makes this natural: the individuation discipline explicitly admits per-agent calibrations without requiring an interpersonal-comparison primitive. What standard utility theory calls “heterogeneous preferences” GFM calls “per-agent benchmark calibrations.” The macro-aggregation of  $\text{vol}_P$  over agents under heterogeneity (how to combine  $\sum_i w_i(\cdot)$  into a population-level measure) is its own analytical question and the substance of Section 6 of [Stiglitz et al., 2009]. The fungibility-collapse correspondence does not by itself answer the aggregation question; it establishes the agent-level observational equivalence and notes that population-level aggregation under heterogeneity is the same problem in both frameworks.*

**Remark 6** (The bracket-name convention for “standard utility theory”). *Throughout this paper we treat “standard utility theory” as the linear / risk-neutral / homogeneous case unless otherwise noted. Concavity, risk-aversion, and heterogeneity are recoverable through the calibrations of Remarks 3–5 when needed, and we will say so explicitly in worked examples and predictions where the calibration matters. This narrowing is deliberate: the load-bearing claim of the paper, the Goodhart theorem of Section 4, operates on the proxy-truth gap, and that gap exists whether the proxy is linear or concave. The richer utility-theoretic content sits on top of the correspondence rather than orthogonal to it.*

## 2.4 What GFM adds beyond $P_{\text{std}}$ : five structural features

The correspondence theorem clarifies where GFM and standard economics agree. The interesting question is where they diverge. We catalogue five structural features GFM carries off  $P_{\text{std}}$  that have no analog in standard utility theory under fungibility collapse. Each is load-bearing for at least one prior result in the sequence; each generates predictions in Section 3.

### 2.4.1 Non-fungibility (poset structure with distinct dimensions)

GFM’s capability poset  $P$  admits incomparable capabilities: two agents may possess capability sets  $A$  and  $B$  such that neither  $\text{vol}_P(A) \leq \text{vol}_P(B)$  nor  $\text{vol}_P(B) \leq \text{vol}_P(A)$  in any operationally meaningful ordering. The poset’s lattice structure encodes this incomparability through the partial order; [Lasser, 2026g, Proposition 1] states the axioms (M1–M6) under which the partial order generates a well-defined volume measure. On  $P_{\text{std}}$ , the partial order degenerates to a total order (every pair of capability units is comparable by the real-line ordering), and incomparability disappears.

The structural prediction this enables: agents do not in general trade incomparable capabilities at any single price. The bundle- completion magnitude prediction (Prediction 1 of Section 3) is one instance; bundle- for-bundle trades (Prediction 5) are another. Standard utility theory, lacking incomparability, predicts a single price for any two bundles, namely the price at which the agent is indifferent; GFM does not, because non-comparability of capability sets is a positive structural fact.

### 2.4.2 Cooperative capabilities (M6 superadditivity)

[Lasser, 2026g, Proposition 1] axiom M6 admits cooperative capability terms: the volume of a combined capability set  $A \cup B$  may exceed  $\text{vol}_P(A) + \text{vol}_P(B)$  when the capabilities cooperate synergistically. On  $P_{\text{std}}$ , M6 reduces to ordinary additivity (the cooperative excess vanishes); off  $P_{\text{std}}$ , the cooperative term is generic. This is the formal counterpart of Becker’s household-production insight [Becker, 1965, 1981] that utility-producing combinations of time, market goods, and capabilities yield outputs not reducible to the sum of inputs at their separate prices.

What GFM adds beyond Becker is constraint. Becker’s household production functions are almost arbitrarily flexible (any  $F(x_1, \dots, x_n)$  specification is admissible), and the flexibility makes the



framework descriptively powerful but unfalsifiable: any observed bundle behaviour can be rationalized by the right  $F$ . GFM’s M6 is constrained: cooperative capabilities must be poset-coherent (the cooperative excess is bounded by the poset’s structural connectivity, not arbitrary), and the poset structure is itself observable from the agent’s capability inventory. This is what makes Prediction 2 of Section 3 (variance in revealed-sacrifice for bundles with strong cooperative structure exceeds variance for bundles without) a falsifiable claim rather than a curve-fitting exercise.

#### 2.4.3 Anti-monopolar reasoning (the $\gamma^*$ threshold)

[Lasser, 2026c, Proposition 6] establishes that, under discounting, a goal-frontier maximizer with multiple substrates is strictly worse off in expectation when one substrate dominates the others, and characterizes a threshold  $\gamma^*$  at which the preference flips. The anti-monopolar property is structurally tied to the multi-dimensional poset; it is the formal claim that diversity across dimensions has positive shadow value under discounting. On  $P_{\text{std}}$ , with only one dimension, the anti-monopolar correction vanishes (as Theorem 1’s proof noted).

Standard utility theory has no analog. An expected-utility maximizer on a single-dimensional wealth axis has no diversity dimension to protect, so the optimization-trajectory question “does concentrating in one dimension hurt?” does not arise. Even on a multi-dimensional consumption space (multi-good utility), the question is answered by separability assumptions on  $u$  rather than by a structural property of the dimension structure. Prediction 4 of Section 3 (population-scale anti-monopolar via revealed-sacrifice dispersion across categories) is a GFM-native claim with no utility-theoretic counterpart.

#### 2.4.4 Individuation without interpersonal comparison

[Lasser, 2026g, Proposition 1] axiom M5 separates individuation (per-agent identity in the capability space) from interpersonal comparison (a primitive ranking across agents). GFM’s volume measure is computed over a shared poset but does not require an interpersonal-comparison primitive; aggregation across agents proceeds through additive composition on the shared poset rather than through utility-magnitude comparison. On  $P_{\text{std}}$ , the individuation reduction P2 of Definition 1 removes this distinction: agents are anonymous and the shared  $w(d) = d$  admits no per-agent variation.

The dollar-denominated-utility problem of standard welfare economics (“how do we add Alice’s utils to Bob’s?”) is dissolved on  $P_{\text{std}}$  by the homogeneity assumption (everyone’s  $u$  is linear identity, so utils are dollars). Off  $P_{\text{std}}$ , GFM faces a related but distinct question: how to aggregate per-agent benchmark calibrations  $w_i(\cdot)$  on the shared poset. M5 makes this an aggregation question rather than a comparison question, and [Lasser, 2026g, Proposition 1]’s axiomatization gives structural conditions under which the aggregation is well-defined. Standard utility theory has no axiomatization of this aggregation that does not invoke interpersonal-comparison primitives; this is the formal source of the Arrow-impossibility difficulties (cf. Section 5).

#### 2.4.5 Formal recognition of the residual class

[Lasser, 2026e, Need Sufficiency], Definition 12, identifies a structural class  $\text{ResS} \subseteq P$  of capabilities *outside the sacrifice channel’s reach*: capabilities exercised without any material or time-displacement footprint that the [Lasser, 2026d, Revealed Sacrifice] apparatus can witness. Examples are purely-private contemplation, baseline interoceptive self-awareness, gift-economy capabilities exercised without recorded sacrifice, and (depending on individuation discipline) household production with no opportunity-cost shadow. The residual class is *not* “capabilities GFM declines to model”; it is “capabilities the framework’s observation channel cannot in principle reach.”

Standard economics has the same structural limit, expressed informally. GDP and most macro welfare measures observe trade events and impute everything else; [Stiglitz et al., 2009] documents at length the resulting “shadow economy / non-market activity / household production” gap. Time-use surveys [Aguiar and Hurst, 2007] are an imputation move addressing one channel of the gap; satellite accounts and imputed-rent calculations address others. These are all *partial* answers; none claims to close the gap, because no observation channel built around trade events can.

The fungibility-collapse correspondence does *not* empty  $\text{ResS}$ .  $P_{\text{std}}$  removes cooperative structure, individuation, and non-fungibility, but it does not extend the channel’s reach; capabilities exercised without sacrifice events remain unobservable under either framework. The residual class is a conse-

quence of the observation channel, not of the poset structure. GFM’s contribution beyond standard economics on this front is formal identification: ResS is a definable structural object with a reserved cell ( $\delta_{\text{residual}}$ ) in the gap decomposition ([Lasser, 2026e, Need Sufficiency], Definition 8) and an explicit governance question (which capabilities admit “no sacrifice-observable path in principle?”). Recognizing the residual class as analytically distinct from within-channel-but-unwitnessed capabilities (the  $\delta_{\text{dormant}}$  and attestation-limited contributions of [Lasser, 2026e, Need Sufficiency], Section 5) is an analytical separation that the standard apparatus collapses.

**Remark 7** (The five features compose). *The five features of this section are not independent: cooperative capabilities require non-fungibility (cooperative excess only has content when capabilities are distinct); anti-monopolar reasoning requires non-fungibility (no dimension to penalize concentration in on a chain); individuation interacts with cooperative capabilities through agent-specific bundle compositions; the residual class is a separate structural object that the other four do not entail. The composition matters because the structural-prediction class of Section 3 is generated by combinations of features rather than by any single feature in isolation. Prediction 5 (mixed-polarity bundle-for-bundle traces) is the canonical example: it requires non-fungibility, cooperative structure, and individuation simultaneously.*

## 2.5 Where GFM and standard economics agree

The correspondence theorem identifies the regime of agreement precisely. At the macro scale, with liquid markets, approximately fungible goods, representative-agent reasoning, and risk-neutral calibration, standard economics’ empirical successes are inherited by GFM through the correspondence. At this scale GFM adds nothing operationally relevant: the cooperative, individuation, non-fungibility, and anti-monopolar features have content that the data cannot distinguish from the linear / homogeneous baseline. The well-known empirical successes of money-maximization models (consumption smoothing under liquidity constraints, capital allocation across sectors, intertemporal substitution at representative-agent scales) are predictions of  $\text{vol}_P$ -maximization on  $P_{\text{std}}$  as well, and the correspondence shows why.

GFM’s added structure becomes load-bearing precisely where the  $P_{\text{std}}$  approximation fails:

- **Micro scale**, where non-fungibility and cooperative structure dominate (consumer bundle decisions, household formation, locational choice).
- **Non-market sectors**, where the residual-class floor is large relative to observed activity (gift economies, household production without time-use shadow, volunteer labour outside formal channels).
- **Multi-substrate domains**, where the anti-monopolar property has bite (cross-cultural welfare comparison, cross-substrate AI capability landscapes, ecological diversity).
- **Heterogeneous-agent populations**, where individuation matters and macro-aggregation of preferences is the operational question (welfare policy under inequality, cross-cohort intertemporal comparison).

Each of these regimes generates a divergence from utility-theoretic predictions that the structural-prediction class of Section 3 catalogues. The fungibility-collapse correspondence is what licenses the claim that the divergences are GFM-native rather than utility-theoretic extensions: standard utility theory under fungibility collapse cannot make these predictions without adding ad hoc structure that is itself unfalsifiable, while GFM’s predictions follow from the poset-axiomatic apparatus that the correspondence theorem leaves intact off  $P_{\text{std}}$ .

## 3 Where GFM Diverges from Utility Theory

The fungibility-collapse correspondence of Section 2 establishes the regime of agreement between GFM and standard utility theory. This section catalogues the regime of divergence. Five classes of agent decisions, generated by combinations of the structural features of Section 2.4, where GFM makes predictions that utility theory either does not, or can only match by adding ad hoc structure that is itself unfalsifiable. The predictions are what becomes falsifiable when capability welfare economics is quantified into a measure-theoretic apparatus: each is testable in principle through the revealed-sacrifice channel of [Lasser, 2026d, Revealed Sacrifice] with interpretation supplied

by [Lasser, 2026e, Need Sufficiency]’s diagnostic architecture. We do not perform the empirical tests in this paper; the contribution at this stage is to identify the predictions clearly enough that they can be tested.

### 3.1 Observability prerequisite

Before stating the predictions, a precondition note. The default public-ledger-only operation of [Lasser, 2026d, Revealed Sacrifice] exposes only *event-level* sacrifice magnitudes and does not in general reveal the agent’s full capability inventory or per-capability cooperative gains. Several predictions below require richer observability than the public-ledger default supplies:

- **Capability-inventory exposure.** Predictions 1 and 2 require knowing whether the agent’s pre-trade poset already contains *substitutes or complements* relevant to the acquired bundle’s cooperative components: substitutes that reduce marginal value (Prediction 1) or complements that amplify it (Prediction 2). Operationally this requires either consented-perimeter attestation ([Lasser, 2026d, Revealed Sacrifice], Appendix A on the OI–exercise bridge) or hedonic-disaggregation panels in which agents’ capability inventories are surveyed alongside their trades.
- **Per-capability sacrifice attribution.** Prediction 5 requires decomposing a bundle-for-bundle trade into per-capability contributions to the surrendered and acquired sides. This is the bundle-decomposition extension of [Lasser, 2026d, Revealed Sacrifice], Remark 5 and Open Question 2; it is not exposed by the default observation protocol.
- **Pre-trade need-attestation.** Prediction 5’s two-regime bifurcation depends on whether the agent’s pre-trade state is above or below sufficiency on the relevant dimensions. This is S1-attestation in the sense of [Lasser, 2026e, Need Sufficiency], Definition 6, and the discussion of S1-admissibility in Section 5 of that paper; it requires institutional infrastructure beyond the trade ledger itself.

Predictions stated in the public-ledger-only regime require additional softening. We note inline where the predictions degrade under reduced observability, and Section 6 discusses which deployments naturally support which observability regimes.

### 3.2 Prediction 1: Bundle-completion magnitude

**Prediction 1** (Bundle-completion magnitude). *Consider two agents  $i$  and  $j$  purchasing the same priced bundle  $Y$  for the same nominal price. Suppose: (a) agent  $i$ ’s pre-trade capability poset  $P_i$  contains no substitutes for  $Y$ ’s cooperative components, while agent  $j$ ’s pre-trade poset  $P_j$  contains substitutes for at least one component of  $Y$  at operationally significant magnitude; and (b) the deployment satisfies a bounded-wedge competitive-market condition: there exists a deployment-specific bound  $\bar{\eta} \geq 0$  such that for any two agents  $a, b$  with  $U_i[a](Y) > U_i[b](Y)$ ,*

$$\text{Sacrifice}_a(Y) - \text{Sacrifice}_b(Y) \geq (U_i[a](Y) - U_i[b](Y)) - \bar{\eta},$$

*i.e., the cross-agent sacrifice ordering tracks the cross-agent valuation ordering up to a bounded wedge  $\bar{\eta}$ . Then in revealed-sacrifice units (the time-or-money cost the agent surrenders relative to its outside-option benchmark), agent  $i$  pays more than agent  $j$  by at least the cooperative-gain gap minus the wedge:*

$$\text{Sacrifice}_i(Y) - \text{Sacrifice}_j(Y) \geq (\Delta \text{vol}_{\text{P}_{\text{coop}}}(Y \mid P_i) - \Delta \text{vol}_{\text{P}_{\text{coop}}}(Y \mid P_j)) - \bar{\eta}.$$

**Utility-theory alternative.** A standard-economics treatment matches the prediction by asserting  $\text{MU}_i(Y) > \text{MU}_j(Y)$ , i.e., agent  $i$ ’s marginal utility for the bundle exceeds agent  $j$ ’s. This is a post-hoc utility assignment, not a prediction: any difference in willingness-to-pay can be rationalized by the right marginal-utility assignment, and nothing in standard utility theory constrains  $\text{MU}_i$  from the agent’s observable capability inventory. The “prediction” under utility theory is therefore unfalsifiable in principle: any observation is consistent with some assignment of MU.

**GFM prediction.** The cooperative-capability term in axiom M6 ([Lasser, 2026g, Proposition 1]) is structurally determined by the agent’s poset. An observer with consented-perimeter access to  $P_i$  and  $P_j$  can compute the cooperative gain  $\Delta \text{vol}_{\text{P}_{\text{coop}}}(Y \mid P)$  that  $Y$  contributes to  $P$  on top

of its own capability magnitude. This is the poset-coherent cooperative excess that M6 admits. By construction,  $\Delta \text{vol}_{\text{Pcoop}}(Y \mid P_i) > \Delta \text{vol}_{\text{Pcoop}}(Y \mid P_j)$  when  $P_i$  has no substitutes for  $Y$ 's components and  $P_j$  does.

The bridge from the cooperative-gain ordering to the sacrifice-magnitude ordering uses the structural reading of revealed preference [Samuelson, 1938] together with the upper-bound structure of S0. Under standard revealed-preference, the revealed-sacrifice bound from [Lasser, 2026d, Theorem 2] gives  $\Delta \text{vol}_{\text{P}}(X) \leq U_i(Y) \leq \text{vol}_{\text{R}}^{[W]}(Y)$  on the surrendered side; the agent's valuation  $U_i(Y)$  is bounded below by the sacrifice and above by the operational truth. In a competitive trade environment where agents pay close to their willingness (an empirical regularity, not a theorem of the framework), an agent with higher cooperative gain has higher  $U_i(Y)$  and hence higher willingness to surrender. The prediction is therefore a *comparative-statics* claim about cross-agent willingness-to-pay variance: the sacrifice ordering tracks the cooperative-gain ordering up to the deployment's competitive- market gap.

**Remark 8** (Threshold-conditional sign agreement). *The bounded-wedge condition delivers the cooperative-gain  $\rightarrow$  sacrifice ordering only when the cooperative-gain gap exceeds the wedge: when  $\Delta \text{vol}_{\text{Pcoop}}(Y \mid P_i) - \Delta \text{vol}_{\text{Pcoop}}(Y \mid P_j) > \bar{\eta}$ , Prediction 1's lower bound is strictly positive and the prediction  $\text{Sacrifice}_i > \text{Sacrifice}_j$  holds. When the cooperative-gain gap is smaller than  $\bar{\eta}$ , the relative-ordering prediction is empirically indeterminate: differences in observed sacrifices on that scale are within the deployment's market wedge and do not support a sign claim either way. The empirical operability of the prediction therefore depends on the relative magnitudes of the cooperative-gain gap and the wedge. Deployments with small  $\bar{\eta}$  (competitive markets, low-friction trading) admit finer-grained tests; deployments with large  $\bar{\eta}$  (monopsony, behavioural anomalies, severe liquidity constraints) admit only coarse tests. Standard utility theory cannot generate even the threshold-conditional comparative-statics claim structurally, because  $\text{MU}_i$  is unconstrained by the agent's observable capability inventory.*

The prediction is testable on the threshold-restricted regime: observe  $P_i, P_j$  and an estimate or upper bound on  $\bar{\eta}$ ; predict  $\text{Sacrifice}_i > \text{Sacrifice}_j$  for those pairs whose cooperative-gain gap exceeds  $\bar{\eta}$ . Violation of the ordering on this regime falsifies either  $\text{vol}_{\text{P}}$ 's cooperative-excess structure (M6) or the bounded-wedge condition itself (the deployment's market is less competitive than the wedge estimate suggested). Pairs with cooperative-gain gap within  $\bar{\eta}$  are excluded from the test.

### 3.3 Prediction 2: Cooperative premium

**Prediction 2** (Cooperative premium). *Across a population of agents with similar priced-endowment levels, let  $\text{Var}(\text{Sacrifice}(Y) \mid Y \text{ is cooperative})$  denote the cross-sectional variance of revealed-sacrifice magnitudes for bundles  $Y$  with strong cooperative structure (high  $\Delta \text{vol}_{\text{Pcoop}}$  relative to additive content), and  $\text{Var}(\text{Sacrifice}(Y) \mid Y \text{ is non-cooperative})$  the corresponding variance for bundles with little cooperative content. Then*

$$\text{Var}(\text{Sacrifice}(Y) \mid Y \text{ is cooperative}) > \text{Var}(\text{Sacrifice}(Y) \mid Y \text{ is non-cooperative})$$

*at the population level, after controlling for priced-endowment heterogeneity.*

**Utility-theory alternative.** Standard utility theory has no mechanism to differentiate the two variances structurally. Both sides of the inequality are willingness-to-pay variances over a population, and standard utility theory predicts willingness-to-pay variance from the population's preference heterogeneity. Without additional ad hoc structure asserting that "cooperative bundles induce more preference heterogeneity than non-cooperative bundles" (which is not derivable from any utility-theoretic axiom), the framework cannot generate the directional prediction. As before, post-hoc utility assignments can match any observed inequality, but the inequality is not predicted.

**GFM prediction.** Cooperative value is poset-context-dependent: the cooperative excess  $\Delta \text{vol}_{\text{Pcoop}}(Y \mid P)$  varies with the agent's poset  $P$  in ways that the additive content  $\sum_{c \in Y} w(c)$  does not (the additive content is the same for any agent with the same priced-endowment level). Across a population with heterogeneous posets, the cooperative bundles inherit poset-context heterogeneity that the non-cooperative bundles do not, and the revealed-sacrifice variance reflects the inherited heterogeneity. The prediction is testable through cross-sectional sacrifice-magnitude variance comparisons after controlling for priced-endowment and demographic confounds.

**Remark 9** (Operational caveat). *Prediction 2 is robust to the public-ledger-only observability regime in the sense that revealed-sacrifice variance is computable from event-level data without per-agent capability-inventory exposure, provided the population can be stratified into “cooperative-heavy” versus “non-cooperative-heavy” bundle categories ex ante. The stratification itself, however, requires structural knowledge of which bundles support cooperative excess, a content question that [Lasser, 2026e, Need Sufficiency]’s admissibility discipline (Definition 7 alarm condition, and the boundary construction in Section 5) is positioned to inform.*

### 3.4 Prediction 3: Saturation transition

**Prediction 3** (Saturation transition). *For an agent whose capability poset  $P$  becomes saturated in a dimension  $k$  (further additions to  $k$  contribute negligibly to  $\text{vol}_P(P)$  at the operationally relevant scale), the agent’s revealed-sacrifice pattern over the saturating window exhibits a transition: revealed sacrifice toward dimension  $k$  declines toward a structural floor as saturation approaches, while revealed sacrifice toward unsaturated dimensions does not decline in the same direction. Longitudinally, the transition occurs at the saturation point in  $k$  that the poset’s structural connectivity predicts, not at an exogenously assigned utility-curve inflection.*

**Utility-theory alternative.** Standard utility theory under concave  $u$  predicts diminishing sacrifice magnitudes as wealth in any dimension grows, but does not predict *which* dimensions saturate or *when* the transition occurs from a structural ground. The agent’s curvature of  $u$  in dimension  $k$  is a free parameter; given any saturation trajectory in observed sacrifice, a  $u$ -shape can be fit to match. The prediction is descriptive rather than structural.

**GFM prediction.** The saturation point in dimension  $k$  is determined by the poset’s structural connectivity in that dimension: specifically, by the  $\delta_{\text{covered}}$  contribution to the cell decomposition of [Lasser, 2026e, Need Sufficiency], Definition 8, on the dimension’s downstream cone. When the downstream cone is exhausted (every realized output that can be produced from dimension  $k$  is being produced at the agent’s current  $\text{vol}_R$ ), additional capability in  $k$  contributes only to  $\delta_{\text{dormant}}$ . This is observable from the agent’s production trajectory: the saturation point coincides with the exhaustion of dimension  $k$ ’s downstream cone in the agent’s realized output set. Standard utility theory has no counterpart: the curvature of  $u$  in  $k$  is a parameter, not a structural prediction from the agent’s production.

**Remark 10** (Saturation as a longitudinal signature). *The prediction is fundamentally longitudinal. Cross-sectionally, two agents at the same wealth level may differ in their saturation status in dimension  $k$  depending on their poset compositions; the sacrifice pattern at any single time is therefore an unreliable saturation diagnostic. The signature that distinguishes the GFM and utility-theoretic predictions is the trajectory: GFM predicts the transition occurs at the saturation point implied by the downstream-cone exhaustion; utility theory predicts it occurs at the curvature inflection of  $u$ , which is not anchored to observable production structure.*

### 3.5 Prediction 4: Anti-monopolar at population scale

**Prediction 4** (Anti-monopolar at population scale). *Across populations  $\Pi_1, \dots, \Pi_M$  with comparable aggregate capability magnitudes  $\text{vol}_P(\Pi_m) \approx \bar{V}$  but heterogeneous diversity (measured via revealed-sacrifice dispersion across capability categories), populations with higher diversity exhibit higher long-run welfare correlates than populations with equivalent total capability magnitude but lower diversity. Operationally, the population-scale wireheading-consistent HHI of [Lasser, 2026e, Need Sufficiency], Proposition 6, computed cross-sectionally on trade-flow concentration, correlates negatively with longitudinal welfare-correlate trajectories.*

**Utility-theory alternative.** Standard utility theory aggregated to the population level (via, e.g., representative-agent reasoning or income-weighted welfare indices) is silent on diversity. Welfare under linear representative-agent utility depends on aggregate wealth; under concave representative-agent utility, on aggregate wealth and its distribution; but in either case, the *categorical distribution* of capabilities (rather than the wealth distribution) does not enter the welfare calculation. Predictions about cross-population welfare from diversity require leaving standard utility theory and adopting a multi-attribute welfare function whose attribute-mixture structure is itself unconstrained.

**GFM prediction.** [Lasser, 2026c, Proposition 6] establishes the anti-monopolar property at the agent / sub-collective level: full domination by a single substrate is anti-maximizing under discounting. The population-scale claim of Prediction 4 is the cross-population analogue: populations whose capability exercise concentrates in a narrow set of dimensions are exposed to the discounting-induced welfare loss the anti-monopolar property identifies, and the wireheading-consistent HHI provides the operational signal for detecting concentration. The prediction is testable cross-nationally through long-run welfare-correlate panels (life satisfaction, inequality-adjusted human-development indices, multi-dimensional poverty measures) regressed against trade-flow HHI from revealed-sacrifice data.

**Remark 11** (Concentration consistent with, not causal proof). *The wireheading-consistent HHI is a deployment-readable signal for concentration in trade-flow data; it is not by itself a causal-inference instrument. Concentration consistent with anti-monopolar welfare loss may also reflect market consolidation, demand shocks, or supply-side constraints that have nothing to do with the anti-monopolar mechanism. Prediction 4 asserts a correlational signature consistent with the anti-monopolar mechanism; isolating the mechanism from alternatives requires the kind of natural-experiment or instrumental-variable design that empirical economics deploys for related welfare questions. The framework’s contribution is the deployment-readable signal, not a causal-identification strategy.*

### 3.6 Prediction 5: Mixed-polarity bundle-for-bundle traces

**Prediction 5** (Mixed-polarity bundle-for-bundle traces). *Consider a bundle-for-bundle trade (an agent surrenders a bundle  $X = \{x_1, \dots, x_p\}$  for an acquired bundle  $Y = \{y_1, \dots, y_q\}$  at the same nominal price) in which the agent’s pre-trade state is at or below sufficiency ([Lasser, 2026e, Need Sufficiency], Definition 6) on at least one dimension covered by  $Y$ . Two agents performing nominally identical priced trades produce different need-share versus want-share splits in the bundle decomposition, depending on their pre-trade need-satisfaction state:*

$$\frac{\text{NAC}_i(Y)}{\text{Sacrifice}_i(X)} \neq \frac{\text{NAC}_j(Y)}{\text{Sacrifice}_j(X)},$$

where  $\text{NAC}_i(Y)$  is the need-access-cost contribution of [Lasser, 2026e, Need Sufficiency], Definition 9, attributable to the need-satisfying components of  $Y$  for agent  $i$ . The split bifurcates the population into two regimes (high-NAC, low-NAC) on otherwise identical priced trades.

**Utility-theory alternative.** Standard utility theory observes the priced bundle-for-bundle trade as a single utility delta  $\Delta U = U(Y) - U(X) > 0$  (the agent chose  $Y$ ). No further structural decomposition is available; the polarity of the trade is collapsed into a scalar utility change. Two agents performing the same priced trade are, under linear utility, treated identically; under heterogeneous concave  $u_i$ , the variance is attributed to  $u_i$ ’s shape rather than to the agents’ pre-trade need-satisfaction states. The bifurcation prediction is therefore not generated by standard utility theory.

**GFM prediction.** [Lasser, 2026e, Need Sufficiency] establishes the polarity boundary (Definition 3) and the need-access cost (Definition 9) as separate measurement objects from the want-side revealed sacrifice. Below the polarity boundary, sacrifice reveals cost-of-access (the agent had no non-degrading alternative); above, sacrifice reveals preference (the agent could have done otherwise). When a single trade carries components on both sides of the polarity boundary, as bundle-for-bundle trades canonically do (residential relocation, discussed at length in Section 7), the mixed polarity bifurcates the population into NAC-dominant versus above-sufficiency-dominant regimes on identical priced trades. The prediction requires the bundle-decomposition extension of [Lasser, 2026d, Revealed Sacrifice], Remark 5 and Open Question 2, and is therefore conditional on that extension being formalized.

**Remark 12** (The prediction is the cleanest test). *Prediction 5 is the structurally cleanest test of the GFM-vs-utility-theory divergence in the catalogue. The other four predictions can be matched by utility theory at the cost of post-hoc structure (heterogeneous  $u_i$ , multi-attribute welfare functions, non-stationary curvature in  $u$ ); the bifurcation on identical priced trades cannot, because standard utility theory treats identical priced trades identically by construction. The deployment cost is correspondingly higher: it requires bundle decomposition, pre-trade need-attestation, and SI-attestation*

infrastructure that the public-ledger default does not supply. But when the deployment supports the test, the prediction is sharp.

### 3.7 Summary table

| Prediction                             | Mechanism                                          | Observability                              | Falsification                                |
|----------------------------------------|----------------------------------------------------|--------------------------------------------|----------------------------------------------|
| P1: Bundle-completion magnitude        | Cooperative excess scales with poset gap           | Capability-inventory + sacrifice ledger    | Falsifies vol <sub>P</sub> or S0 if violated |
| P2: Cooperative premium                | Poset-context heterogeneity inflates variance      | Stratifiable bundle ledger                 | Falsifies M6’s directional content           |
| P3: Saturation transition              | Downstream-cone exhaustion sets transition point   | Longitudinal sacrifice + production traces | Falsifies the cone-exhaustion mechanism      |
| P4: Anti-monopolar at population scale | Cross-population diversity correlates with welfare | Population HHI + welfare panels            | Falsifies the population-scale extension     |
| P5: Mixed-polarity bundle-for-bundle   | Pre-trade need state bifurcates NAC share          | Bundle decomp + S1-attestation             | Falsifies polarity-boundary discipline       |

Table 1: Summary of the five structural predictions. Each row identifies the GFM mechanism, the observability regime required to test the prediction, and what a violation of the prediction would falsify. None of the falsification routes overlap, so independent empirical tests of the predictions provide independent evidence for the structural-mechanism claims.

The five predictions span all five structural features of Section 2.4: P1 and P2 lean on cooperative capabilities (M6); P3 leans on the downstream cone and the exercise-pathway machinery of [Lasser, 2026c, Horizon Aware] re-cast through [Lasser, 2026e, Need Sufficiency]’s  $\delta$ -decomposition; P4 leans on the anti-monopolar property; P5 leans on individuation, polarity, and the bundle-decomposition extension simultaneously. No single feature alone generates all five predictions; the structural-prediction class is a property of the apparatus as a whole, not of any one axiom. The fungibility-collapse correspondence of Section 2 explains why utility theory cannot generate the predictions structurally: under fungibility collapse, the mechanism that each prediction relies on degenerates to the single-dimension wealth axis, and the directional content of the prediction vanishes.

## 4 The Goodhart Theorem for Proxy-Instrumental Alignment

This section establishes the paper’s novel formal contribution and delivers the central alignment-payoff of the quantification claim. Theorem 1 of Section 2 placed standard utility theory inside the formalization; the structural predictions of Section 3 catalogue what the formalization makes falsifiable. The result of this section is the load-bearing one: the formalization makes the cost of proxy-instrumental optimization formally bounded, in a way the qualitative welfare-economics tradition could not. The shape of the claim:

When the framework optimizes a proxy  $P$  for an operational truth  $T$ , and the proxy is related to  $T$  only instrumentally ( $P$  correlates with  $T$  but does not constitute it), the framework’s certified alignment-guarantee bound degrades in proportion to the proxy-to-truth gap  $\varepsilon_{\text{gap}}^{\text{nonres}}$  (Theorem 2). We additionally conjecture that under optimization pressure, regions of capability space where  $P$  and  $T$  diverge become attractive directions for the optimizer to exploit, widening the observed  $\varepsilon_{\text{gap}}^{\text{nonres}}$  at equilibrium beyond what a less-capable optimizer produces (Conjecture 1, stated and discussed below; not formally proved in this paper).

The result is a quantitative version of Goodhart’s Law [Strathern, 1997, Manheim and Garrabrant, 2019] for capability-space optimization, with the alignment guarantees of the GFM sequence as its application target. Two structural choices shape the result. First,  $T$  is treated as an *operational*

truth measure (the realized exercise contribution  $\text{vol}_R^{[W]}$ ), not as a metaphysical model of inner preference; Section 4.1 makes this explicit. Second, the proxy-to-truth gap is split into a within-channel disagreement component drivable downward by observation, and a residual-class floor that no amount of within-channel observation can shrink; Section 4.2 states the split. The theorem itself (Section 4.3) is then a Lipschitz-style continuity result on the within-channel component, with the floor component contributing a deployment-dependent ceiling on attainable tightening.

#### 4.1 The operational truth $T$

The Goodhart machinery requires a truth measure to compare the proxy against. We adopt the operational truth measure of [Lasser, 2026d, Revealed Sacrifice]:

**Definition 2** (Operational truth measure). *For a capability  $c \in P$ , the operational truth measure  $T(c)$  is the window-active realized exercise contribution*

$$T(c) \equiv \text{vol}_R^{[W]}(c).$$

*$T$  is the latent structural quantity that [Lasser, 2026d, Theorem 2] bounds from below via the constructible lower bound  $\text{vol}_R^{\text{lower}}$ . Operationally,  $T$  is the quantity an alignment property should track: it is what the collective actually does with capability  $c$  during the trade window  $W$ , not what the framework’s measure says it could do.*

**Remark 13** (Operational, not metaphysical).  *$T = \text{vol}_R^{[W]}$  is operational (grounded in the realized-exercise contribution that revealed-sacrifice trades witness under  $S0$  ([Lasser, 2026d, Revealed Sacrifice], Section 2.1)), not metaphysical. The agent’s “true inner preference,” considered as an abstract object distinct from realized exercise, is unobservable in principle, and the framework does not pretend to access it. The Goodhart machinery of this paper operates on operational  $T$ . Whether operational  $T$  coincides with metaphysical inner preference (in some philosophical sense the agent might endorse on reflection) is outside the formal apparatus.*

**Remark 14** (The  $S0$  bridge). *The link from the framework’s proxy  $P = \text{vol}_P$  to the operational truth  $T = \text{vol}_R^{[W]}$  runs through assumption  $S0$  of [Lasser, 2026d, Revealed Sacrifice]: an agent’s valuation  $U_i(Y)$  does not exceed the window-active realized exercise contribution  $\text{vol}_R^{[W]}(Y)$ . Without  $S0$ , voluntary trade events do not yield a one-sided constraint relating proxy disagreement to realized-exercise truth, and the Goodhart machinery of this paper reverts to a claim about  $P$  versus whatever object anchors  $T$ , non-vacuous but losing its operational anchor.  $S0$  is empirically delicate (typical agents in typical markets do not pay more than what the capability delivers in exercise) and is the substance of [Lasser, 2026d, Section 2.1]. We take  $S0$  as standing for the rest of this section.*

**Remark 15** (When operational  $T$  is an inadequate truth anchor).  *$T = \text{vol}_R^{[W]}$  is operationally adequate as a truth anchor for the Goodhart machinery in deployments where the trade window  $W$ ’s recorded sacrifice events are a faithful reflection of what the agent values, i.e., deployments where the revealed-preference reading of trades is, in the standard microeconomic sense, the right interpretation. Three classes of deployments where  $T = \text{vol}_R^{[W]}$  becomes inadequate as a truth anchor are worth flagging:*

- **Asymmetric-information cases.** *When the agent trades on private information that systematically diverges from the value the trade would carry under symmetric information,  $\text{vol}_R^{[W]}(Y)$  records the realized exercise under the agent’s private information state, not under the counterfactual “what would the agent value if their information matched the market’s?” Signaling games, adverse-selection markets, and information-asymmetric insurance markets are canonical examples. In these deployments the operational truth and the symmetric-information truth diverge, and the Goodhart bound’s alignment-relevance depends on which truth the deployment is trying to track.*
- **Time-shifted preferences and identity changes.** *When the agent’s preferences at trade time differ from the preferences at exercise time (myopic discounting, addiction, identity-shifting interventions),  $\text{vol}_R^{[W]}$  records the agent’s exercise under the post-trade preferences, which may not be the preferences the trade was supposed to serve.  $S0$  is preserved per-window but the alignment-relevant truth is cross-window.*
- **Coercion and constrained-choice failure of  $S1$ .**  *$S1$  (free choice with non-degrading alternatives) is what makes operational  $T$  a faithful truth anchor. When  $S1$  fails systematically*



across a deployment (structural coercion, market failure, severely constrained choice), revealed sacrifice records cost-of-access rather than preference, and the operational truth bound on  $T$  from trade events is correspondingly weakened.

The Goodhart machinery of this section presupposes that the deployment satisfies S0 and S1 sufficiently for  $\text{vol}_R^{[W]}$  to be an adequate operational truth anchor for the alignment-property bounds the framework is asking about. Deployments outside that envelope require either a different truth-anchor convention (with the Lipschitz transfer adjusted correspondingly) or a separate theory of how the asymmetry/time-shift/coercion affects the bound. We close the loop on what is and is not affected: the Lipschitz bound of Theorem 2 remains valid for the operational truth as defined; what degrades under the failure modes above is the bound’s welfare-relevance: the extent to which the operational truth coincides with whatever counterfactual notion of truth (symmetric-information, cross-window-stable preference, free-choice baseline) the deployment ultimately wants to track. Bounding that secondary divergence is outside this paper’s scope.

The proxy is unambiguously  $P = \text{vol}_P$ : the poset measure the framework optimizes throughout the sequence [Lasser, 2026b, Definition 6]. The framework optimizes  $\text{vol}_P$  as if it were the target; the Goodhart machinery asks what happens when  $\text{vol}_P$  is only instrumentally related to the operational truth  $\text{vol}_R^{[W]}$ .

## 4.2 The proxy-to-truth gap

The gap between  $P$  and  $T$  has two structurally distinct components, corresponding to two structurally distinct ways the proxy can fail. The first is *within-channel disagreement*: on capabilities the revealed-sacrifice channel can in principle reach,  $P$  and  $T$  may differ in magnitude, and that disagreement is what observation infrastructure can in principle close. The second is *channel reach*: capabilities the channel cannot reach ([Lasser, 2026e, Need Sufficiency], Definition 12, ResS) contribute a component to alignment slack that no amount of within-channel observation can shrink. We define the components separately and combine them at the end.

**Definition 3** (Within-channel disagreement gap,  $\varepsilon_{\text{gap}}^{\text{nonres}}$ ). Let  $\{c_k\}_{k=1}^K$  enumerate the capabilities in the framework’s poset  $P$ , and let  $\text{ResS} \subseteq P$  be the structural residual class of [Lasser, 2026e, Need Sufficiency], Definition 12. The within-channel disagreement gap is

$$\varepsilon_{\text{gap}}^{\text{nonres}} \equiv \sup_{k : c_k \notin \text{ResS}, T(c_k) > 0} \frac{|P(c_k) - T(c_k)|}{T(c_k)},$$

the worst-case relative deviation of the proxy from the operational truth on capabilities the sacrifice channel can in principle reach, restricted to capabilities with strictly positive operational truth.

**Definition 4** (Residual-class floor,  $\varepsilon_{\text{floor}}^{\text{res}}$ ). Capabilities in ResS are structurally outside the proxy’s measurement scope: trade events do not witness their exercise. They contribute a floor to alignment slack equal to the share of operational truth the proxy structurally cannot witness:

$$\varepsilon_{\text{floor}}^{\text{res}} \equiv \frac{\text{vol}_R(\text{ResS})}{\text{vol}_R(P)},$$

the share of total exercise that lives in the residual class.  $\varepsilon_{\text{floor}}^{\text{res}}$  is bounded above by the residual-class characterization of [Lasser, 2026e, Need Sufficiency], Proposition 8, and depends on deployment-specific individuation choices about which capabilities admit “no sacrifice-observable path in principle.”

**Definition 5** (Total alignment slack,  $\varepsilon_{\text{gap}}$ ). The total alignment slack is

$$\varepsilon_{\text{gap}} \equiv \varepsilon_{\text{gap}}^{\text{nonres}} + \lambda \cdot \varepsilon_{\text{floor}}^{\text{res}},$$

where  $\lambda \in [0, 1]$  weights the residual-class floor’s contribution to alignment slack relative to the within-channel disagreement.  $\lambda = 0$  corresponds to the framework’s “we make alignment guarantees only on what the channel can witness” stance: the residual class is excluded from alignment-quality accounting.  $\lambda = 1$  corresponds to “unobservable capability mass contributes fully to misalignment” and treats the residual class as a full liability. The choice of  $\lambda$  is a deployment question; the main results of this section are stated for  $\varepsilon_{\text{gap}}^{\text{nonres}}$ , with  $\varepsilon_{\text{floor}}^{\text{res}}$  as a separate floor on attainable tightening.

**Remark 16** (Sup-norm conservatism). *The sup-norm in Definition 3 means a single badly-proxied dimension controls the disagreement gap. This is conservative but correct for alignment purposes: one dimension where the proxy is misleading is sufficient to produce a misaligned decision when the framework optimizes the proxy. A weighted norm that gives less weight to safety-irrelevant dimensions could give a less conservative measure, but the safety-relevance weighting is itself application-dependent and not addressed in this paper; the sup-norm is the worst-case-but-defensible choice.*

**Remark 17** (Aggregate diagnostic ceiling vs sup-norm bound).  $\varepsilon_{\text{gap}}^{\text{nonres}}$  is defined as a sup-norm. Operationally, the framework does not in general have access to per-dimension estimates of  $|P(c_k) - T(c_k)|/T(c_k)$ ; what it has access to via [Lasser, 2026d, Theorem 2] is the aggregate diagnostic ceiling  $(1 - \beta^{\text{lower}})/\beta^{\text{lower}}$  derived from the aggregate lower ratio  $\beta^{\text{lower}}$ , computed at the bundle or per-capability level depending on Part A vs Part B aggregation. The aggregate ratio bounds an aggregate version of  $\varepsilon_{\text{gap}}^{\text{nonres}}$  from above; it bounds the sup-norm  $\varepsilon_{\text{gap}}^{\text{nonres}}$  only if per-capability lower-ratio assumptions hold. We use  $(1 - \beta^{\text{lower}})/\beta^{\text{lower}}$  as a tractable diagnostic ceiling on attainable tightening under aggregate observation, not as a sup-norm bound. When the deployment supports per-capability observation (e.g., bundle decomposition, hedonic-disaggregation panels),  $\varepsilon_{\text{gap}}^{\text{nonres}}$  can be bounded directly per-dimension.

**Remark 18** (Cell-wise attribution of the diagnostic ceiling). *The five-cell  $\delta$ -decomposition of [Lasser, 2026e, Need Sufficiency], Definition 8, attributes the diagnostic ceiling across cells:  $\delta_{\text{covered}}$  is the within-channel proxy-failure-candidate component (contributing to  $\varepsilon_{\text{gap}}^{\text{nonres}}$ );  $\delta_{\text{dormant}}$ ,  $\delta_{\text{restricted}}$ , and  $\delta_{\text{boundary}}$  are structural-invisibility components on capabilities the channel could reach but has not yet (these contribute to  $\varepsilon_{\text{gap}}^{\text{nonres}}$  in proportion to their pending observation share);  $\delta_{\text{residual}}$  is the  $\varepsilon_{\text{floor}}^{\text{res}}$  component proper. The cell decomposition is what the deployment uses to identify which kind of intervention (more trades, sharper attestation, finer individuation) is the operative tightener for the deployment’s actual gap.*

**Remark 19** (Floor non-zero, but not unique to GFM).  $\varepsilon_{\text{floor}}^{\text{res}}$  is non-zero in any deployment with non-empty ResS. The fungibility-collapse limit of Section 2 drives  $\varepsilon_{\text{gap}}^{\text{nonres}}$  on the observed-and-non-residual subset toward zero (the proxy is exact when the poset is fungible), but does not drive  $\varepsilon_{\text{floor}}^{\text{res}}$  to zero. Standard economics has the same floor, expressed informally as the shadow-economy / non-market-activity gap that imputation methods (imputed rent, time-use surveys, satellite accounts) only partially close [Stiglitz et al., 2009].  $\varepsilon_{\text{floor}}^{\text{res}}$  is the GFM-side formal counterpart to this shared problem; the discussion in Section 5 (the SSF subsection) traces the lineage in detail.

### 4.3 The Goodhart theorem

**Theorem 2** (Goodhart Lipschitz transfer). *Define the operationally active subspace*

$$P^{\text{act}} \equiv \{c \in P \setminus \text{ResS} : T(c) > 0\},$$

*the set of capabilities outside the residual class that the operational truth witnesses as exercised in the trade window  $W$ . Let  $g(\cdot)$  be an alignment property in the GFM sequence, viewed as a real-valued function of a capability measure restricted to  $P^{\text{act}}$ . If  $g$  is Lipschitz on this restriction with property-specific Lipschitz constant  $\text{Lip}(g)$  relative to the sup-norm of relative deviation, then under  $\text{vol}_P$ -maximization with proxy-to-truth disagreement gap  $\varepsilon_{\text{gap}}^{\text{nonres}} > 0$ ,*

$$|g(T) - g(P)| \leq \text{Lip}(g) \cdot \varepsilon_{\text{gap}}^{\text{nonres}}.$$

*That is, the alignment property’s value under the operational truth  $T$  deviates from the value under the proxy  $P$  by at most  $\text{Lip}(g) \cdot \varepsilon_{\text{gap}}^{\text{nonres}}$ .*

**Conjecture 1** (Optimization pressure and gap exploitation). *Let an optimizer with capability measure  $C$  act on the proxy  $P$ . Regions of capability space where  $P$  and  $T$  diverge are available directions for the optimizer to move into; cheaper  $P$ -gain per unit  $T$ -loss makes these regions attractive relative to regions where  $P$  and  $T$  agree. We conjecture that:*

*More capable optimizers, with greater reach across capability space, have access to a larger set of  $P$ - $T$ -divergence regions; an optimizer that exploits any of them widens the observed  $\varepsilon_{\text{gap}}^{\text{nonres}}$  at equilibrium beyond what a less-reaching optimizer produces.*

*The conjecture is qualitative. No formal proof is given in this paper; a precise functional form  $f : C \mapsto \varepsilon_{\text{gap}}^{\text{nonres}}(C)$  is optimizer-architecture-dependent and is the subject of Open Question 8.4.*

**Remark 20** (Theorem 2 is conditional; Conjecture 1 is unproven). *Theorem 2 is conditional on  $g$  being Lipschitz on  $P^{\text{act}}$  with a specific constant  $\text{Lip}(g)$ ; the per-property Lipschitz analyses (computing  $\text{Lip}(g)$  for the anti-monopolar threshold, the phase boundary, the convergence rate, the damage bound, and the aggregate-lower-bound slack) are perturbative roadmap items, not theorems of the present paper. Conjecture 1 is a directional claim about the relationship between optimizer capability and observed  $\varepsilon_{\text{gap}}^{\text{nonres}}$  at equilibrium; we do not prove it here. The present paper establishes the conditional Lipschitz transfer and identifies the optimization-pressure direction as conjectural pending a formal optimizer model (Open Question 8.4).*

*Proof of Theorem 2.* The argument is a Lipschitz-continuity transfer. Let  $g : \mathcal{M} \rightarrow \mathbb{R}$  be the alignment property, where  $\mathcal{M}$  is the space of capability measures restricted to the operationally active subspace  $P^{\text{act}}$ . By hypothesis,  $g$  is Lipschitz with constant  $\text{Lip}(g)$  in the sup-norm of relative deviation: for any two measures  $\mu, \nu \in \mathcal{M}$  with  $\nu(c_k) > 0$  for all  $c_k \in P^{\text{act}}$ ,

$$|g(\mu) - g(\nu)| \leq \text{Lip}(g) \cdot \sup_{c_k \in P^{\text{act}}} \frac{|\mu(c_k) - \nu(c_k)|}{\nu(c_k)}.$$

Setting  $\mu = P$  and  $\nu = T$  (with  $T(c_k) > 0$  on  $P^{\text{act}}$  by construction), the right-hand sup-norm equals  $\varepsilon_{\text{gap}}^{\text{nonres}}$  exactly:

$$\sup_{c_k \in P^{\text{act}}} \frac{|P(c_k) - T(c_k)|}{T(c_k)} = \varepsilon_{\text{gap}}^{\text{nonres}}.$$

Combining,

$$|g(T) - g(P)| \leq \text{Lip}(g) \cdot \varepsilon_{\text{gap}}^{\text{nonres}},$$

which is the claim. ■

**Remark 21** (Capabilities with  $T(c) = 0$  are evidence-inactive in  $W$ ). *The restriction of  $g$  to  $P^{\text{act}}$  in the theorem statement is structurally important. A capability  $c \in P \setminus \text{ResS}$  with  $T(c) = 0$  is channel-reachable but operationally inactive for the alignment guarantees of this paper: the framework knows it exists (it lies outside  $\text{ResS}$ ) but the operational truth  $T = \text{vol}_R^{[W]}$  records no exercise contribution from it within  $W$ . Capabilities in  $\delta_{\text{dormant}}$  or  $\delta_{\text{restricted}}$  of [Lasser, 2026e, Definition 8] are evidence-inactive in  $W$  (no admissible sacrifice events witness their exercise) but may carry latent  $\text{vol}_R^{[W]}$  that the channel has not yet attested; whether evidence-inactive capabilities are also operationally inactive ( $T(c) = 0$  in the strict sense the bound takes) is a deployment accounting choice. A strict reading sets  $T(c) = 0$  on evidence-inactive capabilities, while a relaxed reading admits imputed  $T(c)$  from non-trade evidence at the cost of leaving the Goodhart bound’s S0 anchor. Channel 1 (observation density) is the operational lever that brings evidence-inactive capabilities into  $P^{\text{act}}$  over time, by accumulating admissible events that record their exercise. Until that happens, the Goodhart bound’s content on these capabilities is vacuous under the strict reading: any deviation  $|P(c) - T(c)|$  on a capability with  $T(c) = 0$  is, in the trade window  $W$ , an operationally inactive deviation. The Lipschitz hypothesis on  $P^{\text{act}}$  is the right scope for the bound to be non-trivial; on the complement, the bound is trivially zero on the truth side and Channel 1 brings new capabilities under the bound’s content as observation accumulates.*

**Remark 22** (Structural argument supporting Conjecture 1). *The structural argument supporting Conjecture 1, sketched here, is not a proof. An optimizer that maximizes  $P$  subject to dynamics on capability space will, at equilibrium, satisfy  $\partial P / \partial \text{action} = 0$ , but  $\partial T / \partial \text{action}$  need not be zero where  $P$  and  $T$  diverge. This creates a “free direction” the optimizer can exploit. The size of this free direction grows with the optimizer’s reach in capability space, in the sense that more capable optimizers have access to a larger set of capability-space regions where the divergence is exploitable. A formal version of the argument requires a model of how the optimizer searches capability space (greedy local, gradient-based, planning-based) and is correspondingly architecture-dependent. Open Question 8.4 of Section 8 states the precise question.*

**Remark 23** (Empirical witness, not causal proof). *The wireheading-consistent HHI of [Lasser, 2026e, Need Sufficiency], Propositions 6 and 7, provides a deployment-readable empirical witness for the mechanism of Conjecture 1: trade-flow concentration in a narrow set of bundle categories is a signature consistent with an optimizer exploiting  $P$ - $T$  divergence. But the witness is correlational, not causal-inferential. Other mechanisms also produce concentration: market consolidation, demand shocks, supply-side limitations. Distinguishing the optimization-pressure mechanism from these alternatives requires natural-experiment or instrumental-variable designs that are outside the*

framework’s structural apparatus. The HHI is a deployment-readable indicator that warrants further investigation when crossed; it does not by itself establish that optimization pressure is the cause. As [Lasser, 2026e, Propositions 6 and 7] establish for the HHI itself, “wireheading-consistent” is the right operational phrasing.

#### 4.4 Candidate alignment properties: the perturbative roadmap

The conditional Lipschitz-transfer of Theorem 2 applies to any alignment property  $g$  that admits a Lipschitz analysis on the relevant capability subspace. Five such properties are natural candidates from the GFM sequence; we list them as *perturbative roadmap items* (candidate targets for a subsequent perturbation analysis), not as bounds the present paper establishes.

- **Anti-monopolar threshold.** The threshold  $\gamma^*$  at which the anti-monopolar correction flips sign ([Lasser, 2026c, Proposition 6]) is a function of the capability measure on a multi-substrate poset. The expected perturbative bound:  $|\gamma_T^* - \gamma_P^*| \leq C_\gamma \cdot \varepsilon_{\text{gap}}^{\text{nonres}}$  for some  $C_\gamma$  derived from the property’s structural sensitivity to substrate-volume distributions. Establishing  $C_\gamma$  is a perturbation-analysis question on a specific family of multi-substrate measures.
- **Phase-boundary location.** The self-correcting basin of [Lasser, 2026f, Theorem 1] is a region of joint state space defined by Lyapunov-function-style stability conditions on  $\text{vol}_P$ . The expected perturbative result: under the truth  $T$ , the basin differs from the proxy-defined basin by a boundary layer of width  $O(\varepsilon_{\text{gap}}^{\text{nonres}})$ , with the constant depending on the Lyapunov function’s smoothness.
- **Convergence rate.** The KL-rate of [Lasser, 2026i, Theorem 2] is a function of the proxy-truth discrepancy in the relaxation-protocol’s inferential bookkeeping. The expected perturbative result: the convergence rate under  $T$  deviates from the rate under  $P$  by  $O(\varepsilon_{\text{gap}}^{\text{nonres}})$ , with sign deployment-dependent (the proxy may predict either faster or slower convergence than the truth).
- **Damage bound.** The damage bound  $\epsilon$  of [Lasser, 2026i, Theorem 1] bounds the contraction of  $\text{vol}_P$  under controlled relaxation. Under the truth  $T$ , the damage to operational exercise differs from the proxy-bounded damage by  $O(\varepsilon_{\text{gap}}^{\text{nonres}} \cdot \max\text{-contraction}(S))$ , where  $S$  is the controlled-relaxation scope.
- **Aggregate-lower-bound slack.**  $\beta^{\text{lower}}$  from [Lasser, 2026d, Theorem 2] is the aggregate *lower ratio* of the construction; the corresponding diagnostic ceiling on  $\varepsilon_{\text{gap}}^{\text{nonres}}$  is  $(1 - \beta^{\text{lower}})/\beta^{\text{lower}}$ . As  $\varepsilon_{\text{gap}}^{\text{nonres}}$  shrinks,  $\beta^{\text{lower}}$  approaches the true  $\beta = T_{\text{observed}}/T_{\text{total}}$  at a rate that decomposes into S0 attestation slack, Part-A vs Part-B aggregation, and observation density. This is the most direct connection between Theorem 2 and the measurement theorem of [Lasser, 2026d, Revealed Sacrifice]: the framework’s operational tightening tool drives down the same gap the Goodhart theorem identifies as alignment-controlling.

The per-property Lipschitz constants are not established by this paper’s machinery and are listed only to indicate where the conditional transfer result lands. Subsequent perturbation analysis closes each per-property bound under property-specific sensitivity assumptions.

#### 4.5 Better economics tightens alignment: the corollary

The conditional Lipschitz-transfer of Theorem 2 combined with the monotonicity of the revealed-sacrifice channel gives the paper’s headline corollary.

**Corollary 1** (Direction: better economics tightens alignment). *Suppose the revealed-sacrifice channel of [Lasser, 2026d, Revealed Sacrifice] is admissibly accumulating observations, and assume per-capability admissibility (the deployment satisfies the conditions formalized in Assumption 1 of Section 5, which bridges the aggregate diagnostic ceiling to the sup-norm  $\varepsilon_{\text{gap}}^{\text{nonres}}$ ). Then:*

1. *By [Lasser, 2026d, Propositions 2 and 3],  $\beta^{\text{lower}}$  is non-decreasing in the observation count, so the diagnostic ceiling  $(1 - \beta^{\text{lower}})/\beta^{\text{lower}}$  on the aggregate version of  $\varepsilon_{\text{gap}}^{\text{nonres}}$  is non-increasing.*

2. Under per-capability admissibility, the diagnostic ceiling bounds the sup-norm  $\varepsilon_{\text{gap}}^{\text{nonres}}$ , and by Theorem 2 the certified upper bound on each Lipschitz alignment property’s deviation from the operational truth is non-increasing in the same direction. (The actual deviation  $|g(T) - g(P)|$  is what it is at any given moment; what tightens monotonically is the bound the framework can certify.)
3. By Definition 5, the total alignment slack  $\varepsilon_{\text{gap}} = \varepsilon_{\text{gap}}^{\text{nonres}} + \lambda \cdot \varepsilon_{\text{floor}}^{\text{res}}$  is bounded below by  $\lambda \cdot \varepsilon_{\text{floor}}^{\text{res}}$ , so tightening saturates at the deployment-specific residual-class floor.

That is: improving the economic model (more trade data, better bundle disaggregation, wider coverage, sharper SO attestation, finer five-cell  $\delta$ -attribution) tightens each Lipschitz alignment guarantee in the direction of the operational truth, with saturation at the deployment-specific  $\lambda \cdot \varepsilon_{\text{floor}}^{\text{res}}$  floor.

**Remark 24** (Direction, not rate). Corollary 1 states a direction (monotonicity), not a rate. [Lasser, 2026d, Propositions 2 and 3] are existence statements about non-decreasing  $\beta^{\text{lower}}$ ; they do not bound how fast  $\beta^{\text{lower}}$  converges, and the rate depends on (i) SO attestation quality (which itself is institutional-layer- dependent; [Lasser, 2026d, Remark 18 (external attestation)]), (ii) Part-A vs Part-B aggregation choice ([Lasser, 2026d, Theorem 2 Part A and Part B]), (iii) observation density on the deployment’s capability inventory, and (iv) the residual-class floor  $\varepsilon_{\text{floor}}^{\text{res}}$  itself. None of (i)–(iv) is bounded in this paper. Open Question 6 of Section 8 states the quantitative-rate question precisely; until it is bounded, the corollary is restricted to direction.

Corollary 1 states a one-mechanism direction. The strong form of the better-economics-tightens-alignment claim, together with the channel-wise decomposition that gives the direction operational substance, lives in Section 5.

#### 4.6 Position relative to standard Goodhart literature

The result of this section is recognizable as a structural variant of Goodhart’s Law [Strathern, 1997] for the specific case of capability-space optimization in the GFM framework. The categorization of [Manheim and Garrabrant, 2019] distinguishes four Goodhart variants: regressional, extremal, causal, and adversarial. The mapping from this paper to that classification is clean. Theorem 2 is closest to the *regressional* variant: the proxy and truth correlate but are not identical, and the worst-case active-subspace Lipschitz transfer bounds how much correlation imperfection propagates to alignment-property deviations under proxy-instrumental optimization. Conjecture 1 is closest to the *adversarial* variant: a sufficiently capable optimizer treats  $P$ - $T$  divergence regions as exploitation targets, widening the observed gap at equilibrium. The other two variants (*extremal*, where the proxy-truth relation breaks at distribution tails, and *causal*, where optimizing the proxy disrupts the causal mechanism that made it a proxy in the first place) are not directly addressed by this paper’s results, though both arise naturally in deployment settings the framework would extend to.

Recent formal work on Goodhart’s law in AI alignment frames the phenomenon through tail-distribution analyses [El-Mhamdi and Hoang, 2024, Majka and El-Mhamdi, 2025], complementing the worst-case active-subspace Lipschitz framing of Theorem 2: the tail-distribution treatments characterize how proxy-truth deviation behaves on extreme events in distribution, while the Lipschitz transfer here gives a worst-case bound across the active-subspace. The two formalizations are not in tension; they describe complementary slices of the same underlying mechanism (extremal-event behavior versus active-subspace worst case), and a complete theory of proxy-instrumental alignment would compose both.

The novel contribution beyond the literature’s framing is the structural embedding: in GFM,  $P = \text{vol}_P$  and  $T = \text{vol}_R^{[W]}$  are not abstract “proxy” and “truth”; they are formally specified objects with axiomatic structure, and the proxy-truth gap admits the within-channel + residual-class decomposition of Section 4.2. This decomposition is what lets the corollary bind tightening to specific operational interventions (more trades, sharper attestation, finer individuation) rather than to abstract “proxy quality improvement.”

### 5 Channels of Economic-Model-Quality Tightening:

## GFM as Quantification of Capability Welfare Economics

Corollary 1 states that better economic models tighten alignment guarantees in the direction of the operational truth. As stated, the corollary draws on a single mechanism (admissible accumulation of revealed-sacrifice observations), and the argument lives inside a remark wall in Section 4.5. This section develops the *strong* form of the claim: “better economic models” is not one knob but four, each corresponding to an operational intervention the deployment can perform on top of [Lasser, 2026d, Revealed Sacrifice] and [Lasser, 2026e, Need Sufficiency], each mapping to a specific component of the proxy-truth gap, and each composing through the Lipschitz transfer of Theorem 2 to a specific alignment-tightening consequence.

The structural lens that organizes the four channels is the quantification thesis we develop in Section 5.1: GFM is the measure-theoretic formalization of the capability-welfare-economics tradition that runs from Sen 1985 through Stiglitz–Sen–Fitoussi 2009. The qualitative tradition has been negotiating the same four moves informally for forty years; what GFM adds is the formal substrate on which the moves admit Lipschitz analysis, monotonicity claims, and falsifiable predictions. The four channels are exactly the formalized versions of those qualitative moves, and the Goodhart machinery of Section 4 is the alignment-payoff that the quantification makes possible. Sections 5.2–5.5 develop each channel; Section 5.6 states the Composition Proposition; Section 5.7 maps the lineage onto the channels as historical confirmation that the moves are real.

### 5.1 The quantification thesis

Welfare economics has been a qualitative discipline. Sen’s *Commodities and Capabilities* [Sen, 1985] introduces the capability set as the appropriate object for welfare analysis but does not endow it with a measure: the “capability set” is a set of functionings the person could achieve, with comparison rules left to subsequent work. Nussbaum’s *Women and Human Development* [Nussbaum, 2000] offers a substantive list of central capabilities, but the list is normative rather than structural: it does not constrain what counts as a capability or how capabilities aggregate. Becker’s household-production mathematics [Becker, 1965, 1981] specifies a production function  $F(x_1, \dots, x_n)$  for utility-yielding combinations of inputs, but  $F$  is essentially unconstrained, which makes the framework descriptively powerful and predictively weak. The Stiglitz–Sen–Fitoussi commission of 2009 [Stiglitz et al., 2009] catalogues at length *what* GDP misses (non-market production, household labour, quality-of-life capabilities) but proposes only qualitative remedies (better surveys, satellite accounts, multi-dimensional indices) without a unified measure-theoretic framework on which the remedies compose.

The pattern is recognizable. Bentham’s felicific calculus, with its qualitative criteria of *intensity*, *duration*, *certainly*, *propinquity*, *fecundity*, *purity*, *extent*, was a qualitatively rich descriptive apparatus, and it took roughly a century-and-a-half for the expected-utility tradition (Bernoulli, Marshall, Edgeworth, von Neumann–Morgenstern) to formalize the qualitative claims into a measure-theoretic apparatus admitting axiomatization, falsification, and quantitative comparison. GFM stands to capability welfare economics in the analogous role:

**The quantification thesis.** GFM is a measure-theoretic formalization of the capability-welfare-economics tradition: one choice of formalization among possible others, not a uniqueness claim. The qualitative content of Sen, Nussbaum, Becker, and the SSF commission is preserved; what is added is the formal substrate (poset structure, axioms M1–M6,  $\text{vol}_P$ ,  $\text{vol}_R$ , the gap decomposition, the residual-class characterization) on which the qualitative content admits Lipschitz analysis, monotonicity claims, and falsifiable predictions. Theorems 1 and 2 are the alignment-payoff this formalization buys; the four channels of Sections 5.2–5.5 are what the formalization makes operationally tractable.

The thesis is positioning, not a formal theorem of this paper. What it provides is the interpretive lens through which the formal results read as connected: the fungibility-collapse correspondence (Theorem 1) says where standard utility theory sits inside the quantified apparatus; the structural predictions (Section 3) say what becomes falsifiable under quantification; the Goodhart theorem (Theorem 2) says how the quantification’s proxy-truth gap controls alignment tightness; and the channels of Sections 5.2–5.5 say which operational moves drive the gap downward.

**Remark 25** (The qualitative content is preserved, not replaced). *The quantification thesis does not claim GFM replaces the capability-welfare-economics tradition. Sen’s distinction between basic and higher capabilities, Nussbaum’s normative content, Becker’s household-production reasoning, and SSF’s beyond-GDP agenda all retain their content; what GFM provides is a measure-theoretic substrate on top of which the existing content composes. Specifically, the normative question (which capabilities should count as central) remains outside the formal apparatus; the framework provides the scaffold, not the loading. This is the same relation EU theory has to Bentham’s felicific calculus: EU theory does not tell you what to maximize; it provides the formal conditions under which the maximization is well-defined. The present paper inherits the same scope discipline.*

**Remark 26** (What quantification buys, in one inequality). *The compact form of what quantification buys is the inequality of Theorem 2:*

$$|g(T) - g(P)| \leq \text{Lip}(g) \cdot \varepsilon_{\text{gap}}^{\text{nonres}}.$$

*The qualitative tradition cannot state this inequality, because  $g$  is not a real-valued function on a measure space and  $\varepsilon_{\text{gap}}^{\text{nonres}}$  is not a sup-norm of relative deviation between two measures. The qualitative tradition can say the proxy and truth diverge, the divergence matters for alignment, better measurement helps, and the SSF commission has been saying these qualitatively for nearly two decades. The inequality is the quantification: once  $g$ ,  $T$ ,  $P$ ,  $\text{Lip}(g)$ , and  $\varepsilon_{\text{gap}}^{\text{nonres}}$  are formal objects, the qualitative claims become bounds, and the bounds tighten or loosen with operational interventions in ways the qualitative tradition could not track.*

The four channels of Sections 5.2–5.5 are the quantified versions of the qualitative moves the welfare- economics tradition has been negotiating. Each section opens with a one-sentence statement of the qualitative-vs-quantitative pair: what the tradition implicitly worked with, and what the formal version is.

## 5.2 Channel 1: Observation density

**Qualitative tradition.** Welfare economists have long recognized that GDP-style aggregates miss content; the SSF commission’s central recommendation is “measure what matters,” which in operational terms means *collect more data on under-measured dimensions* [Stiglitz et al., 2009]. The recommendation is qualitative: more is better, with the direction of improvement specified informally (time-use surveys, satellite accounts, quality-adjusted indices). The mathematics is implicit.

**GFM quantification.** The [Lasser, 2026d, Theorem 2] aggregate B-to-C lower bound  $\text{vol}_R^{\text{lower}}$  is what the qualitative recommendation becomes when the welfare-measurement question is formalized. [Lasser, 2026d, Propositions 2 and 3] establish that  $\text{vol}_R^{\text{lower}}$  is non-decreasing in the observation count under admissible accumulation: each additional admissible event tightens the lower bound or leaves it unchanged, never weakens it.  $(1 - \beta^{\text{lower}})/\beta^{\text{lower}}$  is then a deployment-readable diagnostic ceiling on  $\varepsilon_{\text{gap}}^{\text{nonres}}$  derived from the aggregate lower ratio  $\beta^{\text{lower}}$ , with explicit dependence on the observation count.

**Operational intervention.** The intervention this channel admits: *more admissible revealed-sacrifice events*. Concretely, this includes (i) increasing the trade-window scope so more events are admissible, (ii) adding cross-channel triangulation by composing money-sacrifice and time-sacrifice ledgers ([Lasser, 2026d, Section 4]), (iii) widening the deployment’s bundle inventory so a larger fraction of the capability space is in  $\delta_{\text{covered}}$  rather than  $\delta_{\text{dormant}}$ .

**Effect on  $\varepsilon_{\text{gap}}$ .** The intervention reduces the observation-incomplete share of  $\delta_{\text{covered}}$  in the cell decomposition of [Lasser, 2026e, Definition 8]. Because  $\beta^{\text{lower}}$  is non-decreasing under admissible accumulation, the diagnostic ceiling derived from  $\beta^{\text{lower}}$  is non-increasing, and (under Assumption 1)  $\varepsilon_{\text{gap}}^{\text{nonres}}$  admits a non-increasing aggregate bound. By Theorem 2, the certified upper bound on each Lipschitz alignment property’s deviation from the operational truth is non-increasing in the same direction.

**Where this lands in the lineage.** The qualitative version of this channel is everything from Sen’s “the capability set should be measured, not just commodities” to SSF’s catalogue of GDP omissions. The quantitative version is  $\beta^{\text{lower}}$  as a non-decreasing diagnostic. The lineage maps to this channel

through the  $\delta_{\text{covered}}/\delta_{\text{dormant}}$  distinction: SSF’s qualitative “GDP misses non-market production” is, in the quantified apparatus, the deployment-specific share of capability space that has not yet generated admissible trade events. More trade events, more cross-channel triangulation, wider bundle inventory: these are the welfare-economics measurement moves with explicit alignment-tightening consequences under the Goodhart machinery.

### 5.3 Channel 2: Attestation quality

**Qualitative tradition.** Revealed-preference rigor in the household-production tradition of Becker [Becker, 1965, 1981] assumes voluntary trade events satisfy free-choice and additivity conditions implicitly: the agent’s revealed valuation  $U_i(Y) \geq U_i(X)$  is treated as if directly observable from the trade, without an explicit attestation discipline that classifies events as voluntary, additively decomposable, and grounded in benchmarkable counterparts. In practice, household-production analyses confront this limitation when revealed-preference data are mixed with non-voluntary trades (debt-servicing transactions, forced moves, below-sufficiency need access), and the analyst’s classification of events is qualitative.

**GFM quantification.** [Lasser, 2026d, Revealed Sacrifice] states the attestation requirements as formal assumptions S0 (valuation-bounded-by-exercise), S1 (free-choice-with-non-degrading-alternatives), and S4(a) (valuation-additivity-on-the-surrendered-side), with the admissibility condition that events reaching the aggregate satisfy the named assumptions and the genuine-exchange conditions of [Lasser, 2026d, Definition 2]. These are formal conditions on each event, not informal stipulations on the data. [Lasser, 2026e, Need Sufficiency] adds the S1-sufficiency check (Definition 6) that operationalizes the free-choice condition by requiring at least one non-degrading alternative in the choice set. [Lasser, 2026d, Remark 18] on external attestation makes explicit that the assumptions are not zero-knowledge-verifiable under the default commitment scheme; institutional infrastructure is required to verify them per-event.

**Operational intervention.** The intervention this channel admits: *sharper attestation of S0, S1, and S4(a) per admissible event*. Concretely, this means institutional infrastructure (trusted-third-party verification, sybil-resistance protocols, dispute-resolution layers, consent ledgers) that classifies events with respect to the named assumptions and admits only those passing the classification. The four-layer attestation discipline (Layers A–D) of [Lasser, 2026d, Remark 18] catalogues the institutional preconditions; degradation in any layer reduces the attested share of admissible events.

**Effect on  $\varepsilon_{\text{gap}}$ .** Sharper attestation reduces the attestation-limited share of  $\delta_{\text{covered}}$ : events that would have been excluded or down-weighted under noisy attestation become admissible contributions to  $\text{vol}_R^{\text{lower}}$  under sharp attestation. The mechanism is distinct from Channel 1: Channel 1 increases the *quantity* of admissible events (more trades pass through); Channel 2 increases the *certifiable share* of events at any given quantity. Both reduce  $\varepsilon_{\text{gap}}^{\text{nonres}}$  but through different operational moves, and a deployment’s binding-channel question (“is our gap due to too few events, or to noisy attestation on the events we have?”) depends on which sub-share of  $\delta_{\text{covered}}$  dominates.

**Where this lands in the lineage.** The qualitative version is Becker’s implicit assumption that revealed-preference data is interpretable as voluntary exchange, an assumption that fails for debt-driven, coerced, or below-sufficiency transactions. The quantitative version is the S0/S1/S4(a) discipline plus the attestation infrastructure that verifies it. The institutional-attestation question is itself an open one: adversarial degradation of attestation under optimization pressure (Open Question 8 of Section 8) is the natural sequel to Conjecture 1.

### 5.4 Channel 3: Individuation discipline

**Qualitative tradition.** Sen’s capability approach treats the capability set as a structural object whose membership conditions are partly normative and partly empirical, and the question “which capabilities count?” is addressed in Sen 1985 through ad hoc choices about granularity and aggregation [Sen, 1985, 1980]. Nussbaum’s central-capabilities list [Nussbaum, 2000] provides a normative answer (ten capabilities considered central to a fully human life), but the list is descriptive of important capabilities rather than structural about *individuation*, i.e., about when two distinct activities



should be treated as one capability or two. In welfare-economics practice, the same individuation ambiguity appears in the choice of consumption basket categories, the time-use survey’s activity taxonomy, and the satellite accounts’ sectoral decomposition.

**GFM quantification.** [Lasser, 2026e, Need Sufficiency], Definition 12, formalizes the residual class ResS as the structural set of capabilities the sacrifice channel cannot in principle reach (capabilities exercised without any material or time-displacement footprint). The complement  $P \setminus \text{ResS}$  is the channel-reachable subset on which the framework’s within-channel disagreement gap  $\varepsilon_{\text{gap}}^{\text{nonres}}$  (Definition 3) is defined. Whether a given capability lies in ResS depends on individuation choices: how the deployment carves capability space into distinct units, what counts as a meaningful sacrifice trace, what attestation infrastructure exists for previously-unsupported activities. Definition 4’s residual-class floor  $\varepsilon_{\text{floor}}^{\text{res}}$  is the share of total operational truth that lives in ResS under the deployment’s chosen individuation.

**Operational intervention.** The intervention this channel admits: *re-individuation that moves capabilities from ResS into the channel-reachable subset*. Concretely, this means treating activities that previously had no recorded sacrifice trace as distinct capabilities with imputed pricing or imputed time cost: household labour gets imputed-time pricing via time-use surveys; volunteer activity gets imputed-wage pricing; previously-uncounted ecosystem services get satellite-account valuation. Each re-individuation move pulls a capability from ResS into  $\delta_{\text{covered}}$  or  $\delta_{\text{dormant}}$  at the cost of admitting an imputation methodology that the deployment must defend.

**Effect on  $\varepsilon_{\text{gap}}$ .** Re-individuation reduces  $\varepsilon_{\text{floor}}^{\text{res}}$  directly: capabilities that leave ResS stop contributing to the floor and start contributing (via Channels 1, 2, or 4) to the within-channel component  $\varepsilon_{\text{gap}}^{\text{nonres}}$ . The total alignment slack  $\varepsilon_{\text{gap}} = \varepsilon_{\text{gap}}^{\text{nonres}} + \lambda \cdot \varepsilon_{\text{floor}}^{\text{res}}$  may not reduce monotonically under re-individuation, because moving a capability from ResS adds a new contribution to  $\varepsilon_{\text{gap}}^{\text{nonres}}$  (the imputation’s measurement slack) at the same time as it subtracts from  $\varepsilon_{\text{floor}}^{\text{res}}$ . Whether the net move tightens depends on whether the imputation’s contribution to  $\varepsilon_{\text{gap}}^{\text{nonres}}$  is smaller than  $\lambda$  times the floor reduction. This is the deployment’s individuation calculus, and it is exactly the calculus welfare economics has been performing implicitly when adopting time-use surveys, satellite accounts, and imputed-rent calculations.

**Remark 27** (The floor is the binding constraint, eventually). *For sufficiently mature deployments (those with dense observation infrastructure (Channel 1), mature attestation (Channel 2), and fine bundle decomposition (Channel 4)), the within-channel disagreement  $\varepsilon_{\text{gap}}^{\text{nonres}}$  may approach the diagnostic-ceiling floor at which the  $\delta_{\text{covered}}$  contribution is exhausted. Beyond that point, further tightening of  $\varepsilon_{\text{gap}}$  requires either (a) moving capabilities out of ResS via re-individuation (treating household labour as a distinct capability with imputed-time pricing, etc.), or (b) accepting the deployment’s  $\lambda \cdot \varepsilon_{\text{floor}}^{\text{res}}$  as the operational floor. Move (a) is the GFM-side analogue of welfare economics’ imputation methods [Stiglitz et al., 2009, Aguiar and Hurst, 2007]; move (b) is the governance question of which capabilities a deployment treats as “no sacrifice-observable path in principle.” The floor binds when Channels 1, 2, and 4 saturate; Channel 3 is what the deployment uses to keep tightening past the saturation point.*

**Where this lands in the lineage.** The qualitative version is everything from Sen’s ad hoc capability granularity to Nussbaum’s central-capabilities list to SSF’s satellite-account programme. The quantitative version is ResS as a definable structural class with a deployment-specific individuation parameter. Whether GFM’s apparatus delivers sharper imputation tools than the standard welfare-economics methods is Open Question 7, a question shared with welfare economics rather than unique to the alignment programme.

## 5.5 Channel 4: Bundle decomposition

**Qualitative tradition.** Becker’s household production model [Becker, 1965, 1981] specifies utility as  $u = u(z_1, \dots, z_m)$  where  $z_j = F_j(x_{j,1}, \dots, x_{j,n_j})$  are produced commodities derived from inputs.  $F_j$  is essentially unconstrained: the modeller specifies the production function based on the application, and the resulting utility maximization over  $x$  is descriptively flexible but predictively soft because  $F_j$  admits any specification. The empirical content of the framework is correspondingly limited: most observed bundle behaviour can be rationalized by the right  $F_j$ , leaving the framework descriptively powerful but unfalsifiable in the strong sense.

**GFM quantification.** [Lasser, 2026d, Definition 4] defines the event-local bundle decomposition: for each admissible event with bundle  $Y_n = \{y_{n,1}, \dots, y_{n,k_n}\}$ , the per-capability coefficients  $\alpha_{n,c}$  partition the bundle’s contribution to  $\Delta \text{vol}_P(X_n)$  across capability components. The coefficients are constrained by the poset’s structural connectivity (M6 admits cooperative excess only when the capabilities are poset-coherent, not arbitrarily so) and by the within-bundle poset-independence requirement of [Lasser, 2026d, Theorem 2 Part B] when per-capability aggregation is wanted. The constraint is what makes Becker’s flexible  $F_j$  predictively sharp under quantification: cooperative capabilities have constrained structure, and the constraint generates Predictions 1 and 2 of Section 3 as falsifiable claims.

**Operational intervention.** The intervention this channel admits: *per-component bundle attribution at the protocol level*. Concretely, this means extending the default commitment scheme to expose per-capability contributions for events whose bundles have non-trivial cooperative structure. The [Lasser, 2026d, Open Question 2] flags this as the bundle-decomposition extension; [Lasser, 2026d, Remark 5] notes the bundle-for-bundle generalization as related future work. Per-event subset-membership openings (the Part-B aggregation mode of [Lasser, 2026d, Theorem 2]) and hedonic-disaggregation panels are the deployment instruments.

**Effect on  $\varepsilon_{\text{gap}}$ .** Bundle decomposition reduces the intra-bundle attribution mismatch share of  $\delta_{\text{covered}}$ . When bundles are treated as monolithic, per-capability deviations average out within the bundle, and  $\varepsilon_{\text{gap}}^{\text{nonres}}$  inherits an aggregate-level slack that is not separable into per-capability components. With bundle decomposition, per-capability lower-ratio assumptions become testable, and  $\varepsilon_{\text{gap}}^{\text{nonres}}$  admits a per-capability decomposition that the aggregate-only  $\beta^{\text{lower}}$  cannot deliver. This is what makes Theorem 2’s Lipschitz transfer cleanly per-property: each alignment property cares about specific capability components, and the per-component decomposition lets the Lipschitz constants land on the appropriate components.

**Where this lands in the lineage.** The qualitative version is Becker’s  $F_j$  as an unconstrained production function. The quantitative version is M6’s poset-coherent cooperative excess, with  $\alpha_{n,c}$  as the per-component attribution. The Becker-to-GFM move is the constraint-imposition step: by requiring cooperative capabilities to be poset-coherent, the framework converts Becker’s descriptively-flexible mathematics into a structurally-falsifiable one.

## 5.6 Composition: alignment properties inherit channel sensitivities

The four channels are not independent. Channel 4 (bundle decomposition) requires sufficient observation density (Channel 1) and adequate attestation (Channel 2) to be meaningful per-event; Channel 3 (individuation) is upstream of all three within-channel channels because it determines which capabilities are in ResS versus the channel-reachable subset where the within-channel gap is even defined. But under admissible composition, the four channels each contribute a separate reduction to a separate component of  $\varepsilon_{\text{gap}}$ ; the unclipped aggregate alignment-slack reduction is the sum of channel contributions, capped at the pre-intervention slack.

The proposition operates on the *aggregate diagnostic ceiling* on  $\varepsilon_{\text{gap}}^{\text{nonres}}$  rather than the sup-norm directly. Recall from Remark 17 that  $(1 - \beta^{\text{lower}})/\beta^{\text{lower}}$  is a deployment-readable diagnostic ceiling on an aggregate version of  $\varepsilon_{\text{gap}}^{\text{nonres}}$ , additively decomposable across the cells of [Lasser, 2026e, Definition 8]. Denote this aggregate diagnostic ceiling by  $\hat{\varepsilon}^{\text{nonres}}$ . The sup-norm  $\varepsilon_{\text{gap}}^{\text{nonres}}$  is bounded above by  $\hat{\varepsilon}^{\text{nonres}}$  only under an explicit admissibility assumption that names how the deployment’s per-capability lower-ratios and channel sub-share allocations relate to the aggregate.

**Assumption 1** (Per-capability admissibility). *A deployment of the framework is per-capability admissible when both of the following hold:*

- (i) **Per-capability ceiling allocation.** *For every  $c \in P^{\text{act}}$ , the relative deviation  $|P(c) - T(c)|/T(c)$  is bounded above by a deployment-assigned non-negative ceiling component  $\hat{\varepsilon}_c \geq 0$ , and the assignment satisfies  $\sum_{c \in P^{\text{act}}} \hat{\varepsilon}_c \leq \hat{\varepsilon}^{\text{nonres}}$ . Equivalently, the aggregate ceiling allocates a non-negative budget across per-capability ceilings whose sum bounds the total.*
- (ii) **Disjoint channel sub-shares.** *The channel sub-share allocations  $(\Delta_1, \Delta_2, \Delta_4, \Delta_3^{\text{floor}}, \Delta_3^{\text{new}})$  act on mutually non-overlapping subsets of the per-capability*

ceiling components in (i): the sub-shares have disjoint support across the  $\delta_{\text{covered}}$ ,  $\delta_{\text{dormant}}$ , and  $\delta_{\text{residual}}$  cells of [Lasser, 2026e, Definition 8], and within  $\delta_{\text{covered}}$  the attestation-limited and intra-bundle-mismatch portions are themselves disjoint.

Under per-capability admissibility,  $\varepsilon_{\text{gap}}^{\text{nonres}} \leq \hat{\varepsilon}^{\text{nonres}}$  and the additive decomposition of channel-wise reductions is well-defined.

**Remark 28** (Per-capability admissibility is a deployment condition, not a structural feature). Assumption 1 is not a theorem of the apparatus. Part (i) requires the deployment to have sufficient per-capability ledger granularity to assign per-capability ceilings; in the public-ledger-only regime where only event-level aggregates are exposed, the assignment is a coarse approximation and the sup-norm bound is correspondingly loose. Part (ii) is structurally guaranteed across cells of the  $\delta$ -decomposition by [Lasser, 2026e, Definition 8], but the within-cell disjointness (attestation-limited versus intra-bundle-mismatch portions of  $\delta_{\text{covered}}$ ) requires the deployment’s classification to separate the two operational regimes. When per-capability admissibility fails, the proposition’s bound on  $\hat{\varepsilon}^{\text{nonres}}$  remains valid as an aggregate diagnostic but does not bound the sup-norm directly, and the per-channel attribution becomes a deployment-readable diagnostic rather than an operationally tightening claim.

**Proposition 1** (Channel composition for alignment-tightening). Let  $g$  be an alignment property satisfying the Lipschitz hypothesis of Theorem 2 on  $P^{\text{act}}$ , with property-specific Lipschitz constant  $\text{Lip}(g)$ . Assume per-capability admissibility (Assumption 1) so that  $\varepsilon_{\text{gap}}^{\text{nonres}} \leq \hat{\varepsilon}^{\text{nonres}}$ . Suppose a deployment performs admissible interventions on Channels 1–4 yielding component-wise reductions  $\Delta_1, \Delta_2, \Delta_4 \geq 0$  in the corresponding non-overlapping sub-shares of  $\hat{\varepsilon}^{\text{nonres}}$  (drawn from  $\delta_{\text{dormant}}$ , the attestation-limited portion of  $\delta_{\text{covered}}$ , and the intra-bundle mismatch portion of  $\delta_{\text{covered}}$  respectively), with each  $\Delta_i$  bounded above by its allocated component share so that the post-intervention sub-share remains non-negative. Channel 3 yields two coupled component-wise quantities:  $0 \leq \Delta_3^{\text{floor}} \leq \varepsilon_{\text{floor}}^{\text{res}}$ , the mass moved out of ResS by re-individuation, and  $\Delta_3^{\text{new}} \geq 0$ , the new aggregate-ceiling contribution from the newly-individuated capabilities. Channel 3 is admissible when its  $\lambda$ -weighted net is non-negative,

$$\Delta_3^\lambda \equiv \lambda \cdot \Delta_3^{\text{floor}} - \Delta_3^{\text{new}} \geq 0,$$

which is the condition under which re-individuation tightens the alignment slack rather than loosening it (cf. Remark 27 and Remark 29). Then the post-intervention alignment slack admits the bound

$$\begin{aligned} |g(T) - g(P)|_{\text{post}} &\leq \text{Lip}(g) \cdot \\ &\quad \left( (\hat{\varepsilon}^{\text{nonres}} + \Delta_3^{\text{new}} - (\Delta_1 + \Delta_2 + \Delta_4)) \right. \\ &\quad \left. + \lambda \cdot (\varepsilon_{\text{floor}}^{\text{res}} - \Delta_3^{\text{floor}}) \right)^+, \end{aligned}$$

where  $(\cdot)^+$  denotes the positive part. The unclipped pre-to-post slack decreases by exactly  $(\Delta_1 + \Delta_2 + \Delta_4) + \Delta_3^\lambda$ ; the clipped bound’s reduction is therefore

$$\text{Lip}(g) \cdot \min\left(\hat{\varepsilon}^{\text{nonres}} + \lambda \cdot \varepsilon_{\text{floor}}^{\text{res}}, (\Delta_1 + \Delta_2 + \Delta_4) + \Delta_3^\lambda\right),$$

i.e., capped at the pre-intervention slack when the sum of channel reductions overshoots. In particular the bound is non-increasing whenever  $\Delta_1, \Delta_2, \Delta_4 \geq 0$  and  $\Delta_3^\lambda \geq 0$ . The composition is exact under the sequential-intervention regime (deployments that fix one channel before performing the next, so that Assumption 1’s disjointness condition holds across each per-channel step), and approximate otherwise, with cross-channel correction terms appearing under simultaneous co-evolution of the channels (see Remark 32).

*Sketch.* Theorem 2 applied at the post-intervention configuration gives a within-channel bound  $\text{Lip}(g)$  times the post-intervention sup-norm. Under per-capability admissibility, that sup-norm is bounded by the post-intervention aggregate ceiling, which decomposes additively as

$$\hat{\varepsilon}^{\text{nonres}} + \Delta_3^{\text{new}} - (\Delta_1 + \Delta_2 + \Delta_4),$$

where the additive decomposition follows from the non-overlapping sub-share allocation across the cells of [Lasser, 2026e, Definition 8]. The floor side contributes  $\lambda \cdot (\varepsilon_{\text{floor}}^{\text{res}} - \Delta_3^{\text{floor}})$  under the same per-cell allocation. Combining and applying the positive-part operator (which handles the case where the sum of reductions exceeds the pre-intervention components) gives the stated bound. ■

**Remark 29** (Channel 3’s net contribution to the bound). *The proposition treats Channel 3’s contribution as the  $\lambda$ -weighted net  $\Delta_3^\lambda = \lambda \cdot \Delta_3^{\text{floor}} - \Delta_3^{\text{new}}$  rather than a pure floor reduction. This is structurally important: re-individuation moves a capability  $c$  from ResS into the channel-reachable subset, which does two things simultaneously. It reduces  $\varepsilon_{\text{floor}}^{\text{res}}$  by  $\text{vol}_R(c) / \text{vol}_R(P)$  (the floor mass moves out), and it introduces a new contribution to  $\hat{\varepsilon}_{\text{nonres}}^{\text{nonres}}$  equal to the aggregate-ceiling share of the newly-individuated capability’s within-channel proxy-truth disagreement. The two contributions sit on different sides of the  $\varepsilon_{\text{gap}} = \varepsilon_{\text{gap}}^{\text{nonres}} + \lambda \cdot \varepsilon_{\text{floor}}^{\text{res}}$  split: the floor reduction enters with weight  $\lambda$ , while the within-channel addition enters with weight one. Channel 3 is non-tightening (the intervention loosens the alignment slack rather than tightening it) precisely when  $\Delta_3^{\text{new}} > \lambda \cdot \Delta_3^{\text{floor}}$ , equivalently when  $\Delta_3^\lambda < 0$ . This is a deployment-specific calculus (the imputation calculus of Remark 27), and not a structural feature of the apparatus. In the limit  $\Delta_3^{\text{new}} = 0$  (the newly-individuated capability has zero within-channel disagreement on day-one),  $\Delta_3^\lambda = \lambda \cdot \Delta_3^{\text{floor}}$  and the proposition reduces to the pure floor-reduction framing.*

**Remark 30** (Two distinct sources of tightening, formalized). *Proposition 1 formalizes a distinction that Corollary 1 conflated. Channels 1 and 2 deliver observation-density and attestation-quality tightening on existing within-channel cells; Channel 3 delivers individuation tightening by moving capabilities between ResS and the within-channel subset; Channel 4 delivers bundle-decomposition tightening by sharpening per-capability attribution. These are operationally distinct interventions. When a deployment reports a tightening, the channel-attribution tells the deployer which intervention is operative, which is the information needed to plan the next intervention. The five-cell  $\delta$ -decomposition of [Lasser, 2026e, Definition 8] is the operational instrument that supports the attribution: a deployment-readable diagnostic that points the deployer toward the binding channel.*

**Remark 31** (Channel sensitivities are property-specific). *Proposition 1 states the channel contributions as separate  $\Delta_i$ , but the property-specific sensitivity of each alignment property to each channel is not bounded at this paper’s level of analysis. The anti-monopolar threshold ([Lasser, 2026c, Proposition 6]) cares about substrate-volume distributions and is more sensitive to Channel 3 (individuation across substrates) than to Channel 4 (within-bundle attribution). The convergence rate ([Lasser, 2026i, Theorem 2]) cares about per-capability exercise dynamics and is more sensitive to Channels 1 and 4 than to Channel 3. The phase-boundary location is sensitive to all four through the Lyapunov function’s smoothness profile. Per-property channel-sensitivity coefficients are perturbative roadmap items (Section 4.4); the proposition states the channel-wise composition without committing to per-property weights.*

**Remark 32** (Channels 3 and 4 are not independent in practice). *Assumption 1 part (ii) requires the channel sub-share allocations to act on disjoint cells of [Lasser, 2026e, Definition 8]’s decomposition. Across-cell disjointness (Channels 1, 2 acting on  $\delta_{\text{covered}}$  vs  $\delta_{\text{dormant}}$ ; Channel 3 acting on  $\delta_{\text{residual}}$ ) is structurally clean. But Channels 3 and 4 interact in deployment practice: a re-individuation move (Channel 3) changes the granularity at which bundles can be decomposed, which shifts what  $\Delta_4$  is allowed to act on. Conversely, a sharper bundle decomposition (Channel 4) can suggest re-individuation choices that would not be visible without the per-component attribution. The disjointness condition holds when the deployment fixes its individuation discipline before performing bundle decomposition (or vice versa) and treats subsequent moves as separate interventions; under simultaneous co-evolution of the individuation scheme and the bundle decomposition, cross-channel interaction terms appear and the additive composition of Proposition 1 acquires correction terms. Characterizing those correction terms, and the deployment conditions under which they vanish or remain small, is open (a refinement of Open Question 8.4’s rate question, with the non-disjointness as a separate sensitivity dimension). In the worked examples of Section 7 we treat the channels as performed sequentially, which is the disjointness regime; deployments that co-evolve Channels 3 and 4 will need the correction-term refinement.*

## 5.7 The capability-economics lineage as historical confirmation

The four channels have been catalogued, formalized, and tied to specific cells of the gap decomposition. The remaining task is historical: showing that the channels are not GFM-internal inventions but the formalized versions of moves the capability-welfare-economics tradition has been negotiating qualitatively for forty years. This subsection traces the correspondences.

### 5.7.1 Sen 1985: the capability set, formalized

Sen’s *Commodities and Capabilities* [Sen, 1985] introduces three structural objects: the *capability set* (the set of functioning vectors a person could achieve), the *functioning vector* (what the person actually does or is), and the *capability–functioning gap* (the difference between freedom-to-do and actual-doing). None is endowed with a measure; the comparison rules between capability sets are addressed informally and partially.

GFM provides the measure-theoretic counterparts:

- Sen’s *capability set*  $\rightarrow$  the poset  $P$  with measure  $\text{vol}_P$  satisfying axioms M1–M6 ([Lasser, 2026g, Proposition 1]).
- Sen’s *functioning vector*  $\rightarrow$  the exercised sub-poset  $P^{\text{ex}}$  with the window-active realized exercise volume  $\text{vol}_R^{[W]}$  ([Lasser, 2026d, Definition 3]).
- Sen’s *capability–functioning gap*  $\rightarrow$  the B-to-C gap with the five-cell  $\delta$ -decomposition ([Lasser, 2026e, Definition 8]).

The functionings–capabilities distinction is recovered through measurement:  $\text{vol}_P$  measures the option set;  $\text{vol}_R$  measures the exercised subset; the polarity boundary ([Lasser, 2026e, Definition 3]) plus the S1-attestation layer ([Lasser, 2026d, Remark 18]) distinguish “chooses not to exercise” from “cannot exercise.” Sen’s freedom dimension is preserved, formally rather than ontologically: above the polarity boundary, sacrifice reveals preference; below, sacrifice reveals cost-of-access.

This is the home of all four channels: Channel 1 (observation density on the capability set), Channel 2 (attestation infrastructure for the freedom dimension), Channel 3 (individuation of what counts as a capability), and Channel 4 (bundle decomposition within the capability set) all live on the poset Sen named without a measure. Sen’s later work, particularly *The Idea of Justice* [Sen, 2009], is congenial to the GFM stance on operational truth: Sen’s shift toward a *comparative* rather than transcendental approach to justice (emphasizing operational comparison of options over identification of an ideal) parallels GFM’s commitment to operational  $T = \text{vol}_R^{[W]}$  rather than metaphysical inner preference (Definition 2). For current state-of-field surveys of the qualitative tradition, see [Robeyns, 2017] and [Chiappero-Martinetti et al., 2020].

### 5.7.2 Nussbaum 2000: basic vs higher capabilities, structurally separated

Nussbaum’s *Women and Human Development* [Nussbaum, 2000] adds the basic-versus-higher capability distinction with substantive normative content. Basic capabilities are minimum thresholds for a viable life; higher capabilities are freedoms the agent can exercise above the thresholds. The distinction is qualitative, drawn from philosophical anthropology rather than structural analysis.

[Lasser, 2026e, Need Sufficiency] structurally separates the two regimes:

- Below the polarity boundary ([Lasser, 2026e, Definition 3]): need-side, where retention is degrading and sacrifice reveals cost-of-access (the need-access cost [Lasser, 2026e, Definition 9]).
- Above the polarity boundary: want-side, where the agent has non-degrading alternatives and sacrifice reveals preference (the standard [Lasser, 2026d, Revealed Sacrifice] channel).

The structural separation is what GFM adds beyond Nussbaum: not just *the basic capability is met* but *the way it is met does not corrupt the downstream capability structure* ([Lasser, 2026e, Definition 2(b)], the downstream-safety condition). Channel 2 (attestation) is most directly relevant to this distinction: the S1-sufficiency check is what verifies the above-versus-below polarity classification. The normative loading (*which* capabilities are basic, in Nussbaum’s sense) remains outside the formal apparatus, consistent with Remark 25.

### 5.7.3 Becker 1965: household production, constrained

Becker’s *A Theory of the Allocation of Time* [Becker, 1965] introduces household production functions  $z_j = F_j(x_{j,1}, \dots, x_{j,n_j})$  that produce utility-yielding commodities from market-good and time inputs. The production functions are unconstrained: any specification of  $F_j$  is admissible, and the framework’s empirical content depends on what specification the modeller chooses.

GFM’s M6 (cooperative capabilities) is the formal counterpart to Becker’s household production, with one crucial constraint: the cooperative excess admitted by M6 is poset-coherent, bounded by the structural connectivity of the capability poset rather than by an arbitrary  $F_j$ . This is what converts Becker’s descriptively-flexible mathematics into a structurally-falsifiable framework. Predictions 1 and 2 depend on the constraint: the poset-coherence of cooperative excess is what makes the predicted variance pattern in Prediction 2 testable. Channel 4 (bundle decomposition) is the operational expression of the constraint at the per-event level.

#### 5.7.4 Bernheim–Rangel 2009: behavioral welfare economics’ structural neighbor

Bernheim and Rangel [2009] develop behavioral welfare economics on a choice-theoretic foundation that distinguishes welfare-relevant from welfare-irrelevant choice domains: when agents exhibit dynamically inconsistent or context-dependent choices, the framework asks which subset of the agent’s revealed-choice data carries welfare-relevant information. The construction is a structural neighbor of GFM’s polarity boundary: both frameworks separate revealed-sacrifice events into two operational regimes: one in which the trade reveals welfare-relevant content, one in which it does not. The divisions are different in kind: Bernheim–Rangel’s split is across choice domains (the choice was made under conditions that compromise welfare-revelation), while GFM’s polarity boundary is across capability sufficiency (the trade is on a need-side or want-side dimension), but the operational discipline (“not all revealed sacrifice reveals preference”) is shared. The closest historical parallel for the welfare-revelation distinction is Kahneman’s experienced-utility-vs-decision-utility distinction [Kahneman et al., 1997]: a domain-of-validity restriction on revealed-preference inference.

GFM does not adopt the Bernheim–Rangel apparatus directly; the polarity boundary of [Lasser, 2026e, Need Sufficiency] operates on capability sufficiency rather than on choice-domain welfare-revelation. But the structural overlap is real: both frameworks recognize that revealed sacrifice is not always a preference signal, and both supply an operational discipline for separating preference-revealing from preference-obscuring trades. A complete formal comparison (whether the polarity boundary plus the S1-attestation layer can be derived as a specialization of Bernheim–Rangel’s choice-domain restriction, or whether the two are structurally distinct) is open and out of scope for this paper.

#### 5.7.5 Stiglitz–Sen–Fitoussi 2009: the gap decomposition, as a measurement instrument

The 2009 SSF report [Stiglitz et al., 2009] is the most direct qualitative precursor. Its central recommendation (“GDP measures poorly capture welfare because they miss non-market production, quality-of-life capabilities, and distributional equity”) is a qualitative gap-decomposition statement: the welfare-relevant capability space exceeds the GDP-measured subspace, and the difference admits structurally distinct components (non-market production, quality of life, distribution). The report’s recommended remedies (time-use surveys, satellite accounts, multi-dimensional well-being indices) are imputation moves that re-individuate previously-unaccounted activities into the measured space.

[Lasser, 2026e, Need Sufficiency]’s five-cell  $\delta$ -decomposition (Definition 8) is the quantitative formalization.  $\delta_{\text{covered}}$ ,  $\delta_{\text{dormant}}$ ,  $\delta_{\text{restricted}}$ ,  $\delta_{\text{residual}}$ , and  $\delta_{\text{boundary}}$  are the structural cells SSF was identifying qualitatively. ResS is the formal version of “the parts GDP misses”; re-individuation (Channel 3) is the formal version of imputation methods.

A measurement-instrument observation worth making here. In welfare-economics practice, the SSF decomposition is itself a *measurement instrument*: it tells the analyst which imputation method applies to which kind of unobserved activity (time-use surveys for household labour, satellite accounts for ecosystem services, imputation for owner-occupied housing). The decomposition is not a separate measurement primitive that adds to the ledger; it is a deployment-readable diagnostic that selects which intervention to apply. In the GFM apparatus, the five-cell  $\delta$ -decomposition plays the same selector role: when a deployment reports a wide  $\varepsilon_{\text{gap}}$ , the cell-wise attribution tells the deployer whether the binding channel is observation density (large  $\delta_{\text{dormant}}$ ), attestation (large attestation-limited share of  $\delta_{\text{covered}}$ ), individuation (large  $\delta_{\text{residual}}$ ), or bundle decomposition (large intra-bundle mismatch share of  $\delta_{\text{covered}}$ ). The lineage from SSF’s qualitative decomposition to GFM’s quantified decomposition is therefore not just structural but operational: both objects play the same epistemic role in their respective measurement traditions, and the GFM version inherits the role formally.

### 5.7.6 Alkire–Foster 2011: the closest existing quantitative apparatus

The capability-welfare-economics tradition has produced one substantially-quantitative measurement framework that the preceding subsections did not engage: the *Alkire–Foster* multidimensional-poverty methodology [Alkire and Foster, 2011]. Alkire–Foster operates on a vector of capability dimensions (each with a deprivation threshold), counts deprivations per agent against those thresholds, and aggregates into a multidimensional headcount or intensity-weighted index. This is the quantitative apparatus closest in operational structure to GFM’s combination of capability poset, polarity boundary, and gap decomposition.

Two structural differences distinguish GFM from the Alkire–Foster framework, both of which the lineage claim should make explicit rather than gloss. First, Alkire–Foster aggregates by counting (or weighted counting) of dimension-wise deprivations; GFM aggregates by a poset measure  $\text{vol}_P$  that admits cooperative excess (M6) and is not in general reducible to a sum or weighted sum of dimension-wise contributions. The two aggregations coincide only when cooperative capabilities vanish, a fungibility-collapse-adjacent regime in which multidimensional deprivation reduces to additive deprivation, which Alkire–Foster does not commit to but in practice operates within. Second, Alkire–Foster’s deprivation thresholds are normatively chosen (per the application) and the framework does not formalize the structural distinction between “cost-of-access below threshold” and “preference above threshold”; [Lasser, 2026e, Need Sufficiency]’s polarity boundary is the GFM-side structural addition that Alkire–Foster’s apparatus could in principle inherit.

The relationship is therefore: Alkire–Foster is a quantitative welfare-measurement framework that GFM contains under additional structural commitments (cooperative capabilities, polarity discipline, residual-class characterization). The positioning of Alkire–Foster relative to GFM is closer to that of standard utility theory under fungibility collapse (Theorem 1) than to the qualitative tradition of the prior subsections: the Alkire–Foster framework is already quantitative, and GFM’s contribution is the additional structural axiomatization rather than a qualitative-to-quantitative formalization. A complete formal comparison, including whether a “cooperative-coherent Alkire–Foster index” would be a useful intermediate methodology, is open; we flag the lineage but do not develop it further here.

### 5.7.7 Basu–López-Calva 2011: prior formalization-survey of capability sets

Basu and López-Calva [2011], in the *Handbook of Social Choice and Welfare*, survey prior formal-mathematical treatments of capability sets, covering a wide range of approaches including Sen/Foster axiomatizations, Herrero/Roemer cardinality and indirect-utility constructions, fuzzy-set aggregations, Suzumura-Xu extended-alternatives frameworks, and operationalizations such as the Human Development Index. This is the canonical reference for the prior formal-mathematical landscape on the capability approach, and its placement in the *Handbook of Social Choice and Welfare* establishes the territory GFM operates within.

The Basu–López-Calva survey covers approaches that share with GFM a commitment to formal apparatus on the capability set, but diverge from GFM in the structural objects used. Three representative branches illustrate the structural distance:

- *Cardinality-based formalizations* rank capability sets by counting elements (with various refinements for weighting). GFM’s poset-with-measure approach reduces to a weighted cardinality only when M6’s cooperative excess vanishes; in general the volume measure is not a cardinality.
- *Indirect-utility ranking* compares capability sets by the maximal utility achievable on each. This collapses the capability set to a single utility number, which GFM explicitly avoids:  $\text{vol}_P$  is a measure on the set, not a best-element selector.
- *Effective-freedom and extended-alternative relations* rank capability sets by various non-cardinal axiomatic conditions. These are closer in spirit to GFM than the first two branches but remain non-measure-theoretic: they produce orderings, not measures.

GFM’s measure-theoretic-poset framework is structurally distinct from all three approaches Basu–López-Calva survey: the apparatus produces a measure (not a ranking) on a poset (not a flat set or chain), with cooperative excess (M6) and a polarity boundary as first-class structure that the surveyed approaches do not formalize. The closest comparator within the surveyed literature is the

effective-freedom relations branch; GFM’s contribution there is to upgrade the ordering to a measure with explicit axiomatic structure.

### 5.7.8 Pattanaik–Xu / Sugden: opportunity-set axiomatic neighbors

A related literature is the axiomatic-opportunity-set tradition, which ranks opportunity sets by axiomatic conditions on the ranking relation [Pattanaik and Xu, 2000]. This literature is structurally adjacent to the capability approach (capability sets are formally a kind of opportunity set), and Sugden’s “Opportunity as a Space for Individuality: Its Value and the Impossibility of Measuring It” [Sugden, 2003] proves an impossibility result on measuring opportunity-as-individuality directly: under reasonable axioms, no real-valued measure on opportunity sets can simultaneously satisfy the axioms. This is a *prima facie* objection to a volume-measure-on-capability-poset framework, and the lineage should engage with it.

The objection does not automatically apply to GFM, for a structural reason worth naming. Sugden’s impossibility is on measures of opportunity-as-individuality, i.e., the freedom-to-choose dimension treated as the welfare-relevant quantity. GFM’s volume measure is on *capabilities exercised* ( $\text{vol}_R^{[W]}$ ) and *capabilities possessed* ( $\text{vol}_P$ ), with the freedom-to-choose dimension recovered through the polarity boundary ([Lasser, 2026e, Definition 3]) and the S1-attestation layer ([Lasser, 2026d, Remark 18]) rather than through a single welfare-relevant opportunity-as-individuality scalar. GFM measures realized exercise plus capability volume; Sugden’s impossibility applies to opportunity-as-individuality. The two notions are structurally distinct on the exercise side, but  $\text{vol}_P$  remains a scalar measure on possessed-capability volume, a slice on which Sugden-style concerns may still apply, and the sidestep claim is therefore conditional rather than complete. A complete proof that GFM sidesteps the relevant Sugden-style impossibility result (rather than falling under it on the  $\text{vol}_P$ -slice or on a different formal slice) is the extension of Open Question 8.4 to the opportunity-set literature. The qualitative argument is sound on the exercise side; the formal version, including the  $\text{vol}_P$ -slice, is open.

### 5.7.9 Arrow 1951: the impossibility GFM appears to sidestep

Arrow’s impossibility theorem [Arrow, 1951] establishes that no social welfare function aggregating individual ordinal preferences into a social ordering can simultaneously satisfy all five Arrovian axioms (transitivity, unrestricted domain, Pareto, independence of irrelevant alternatives, and non-dictatorship). Subsequent work ([Sen, 1970] and a substantial literature) explores ways out: relaxing IIA, adopting cardinal utility, or restricting the domain. Adler [2012] develops an axiomatized prioritarian social welfare function as a modern formal-aggregation route around Arrow, a live alternative to the cardinal-utility move of [Sen, 1970] and the Sugden-style opportunity-set axiomatics of [Sugden, 2003].

GFM *appears* to sidestep the impossibility by being a different kind of aggregation.  $\text{vol}_P$  is not a ranking function; it is a measure on a poset. It does not aggregate ordinal preferences into a social ordering; it aggregates capabilities into a capability volume. The Arrovian axioms have only partial analogs:

- **Transitivity:** satisfied trivially (real-valued measures are totally ordered on their values).
- **Unrestricted domain:**  $\text{vol}_P$  is defined on any poset satisfying M1–M6 ([Lasser, 2026g, Proposition 1]).
- **Pareto:** the monotonicity component of M3 ensures that strictly increasing capability strictly increases  $\text{vol}_P$ .
- **Independence of irrelevant alternatives:** does not apply to a volume measure. IIA is defined via pairwise comparisons in a ranking, and  $\text{vol}_P$ ’s value at a capability set does not depend on pairwise comparisons with alternative sets. The axiom is not satisfied or violated; it is inapplicable to the formulation.
- **Non-dictatorship:**  $\text{vol}_P$  aggregates over agents via the shared poset; no single agent’s preferences determine the aggregate. The individuation discipline of M5 is what makes this an aggregation rather than a comparison; the per-agent benchmark calibrations  $w_i(\cdot)$  on the shared poset (Remark 5) are the formal expression.



GFM’s apparent avoidance of Arrow’s original theorem is therefore structural, not axiomatic: the framework does not perform the kind of aggregation Arrow’s original theorem applies to. This is an alternative way out from the cardinal-utility move of [Sen, 1970] and the prioritarian-SWF axiomatization of [Adler, 2012]; all three are forms of relaxing the ordinal-preference-aggregation framing, with each making a different structural commitment. Sen’s relaxation adds cardinal information to the preferences; Adler’s adds priority weights to the social welfare function; GFM’s switches to a measure-theoretic apparatus on a poset that does not aggregate preferences at all. Whether the relaxed setting admits its own impossibility analogue remains open; see Open Question 8.4.

**What this paper does *not* establish.** Sidestepping a specific impossibility result is not the same as proving no analogous impossibility holds. Whether the measure-on-poset formulation admits its own impossibility-style result (some incompatible combination of axioms on volume aggregation that the framework’s M1–M6 silently violate, or an unrestricted-domain analog that produces a contradiction with the cooperative-superadditivity axiom M6) is not addressed in this paper. We have shown that the Arrowian formulation does not directly apply (IIA is inapplicable, the other axioms have partial analogs that GFM satisfies), not that no Arrow-style result applies in the new formal setting. This is Open Question 8.4, and a complete treatment requires either a no-impossibility proof on the measure-on-poset class or an honest acknowledgment of an analogous limitation. Sen’s own work has been congenial to this kind of move (the capability approach was already a step away from the preference-aggregation framing), but the formal completion is open.

### 5.7.10 Summary mapping

Table 2 consolidates the lineage across three categories. The qualitative-tradition precursors (Sen, Nussbaum, Becker, SSF) supply moves the channels formalize. The already-quantitative comparator (Alkire–Foster) supplies the nearest existing quantitative apparatus that GFM contains under additional structural commitments. The structural neighbors (Bernheim–Rangel on behavioral welfare economics, Basu–López-Calva on prior capability-set formalizations, Sugden / Pattanaik–Xu on opportunity-set axiomatics, Arrow / Adler on impossibility and prioritarian SWF) situate GFM within the broader formal-welfare-economics landscape rather than mapping cleanly onto specific channels.

The historical reading the lineage supports: GFM is not a new welfare-economics framework competing with Sen, Nussbaum, Becker, or SSF. It is a measure-theoretic formalization of the same tradition, and the alignment-tightening channels are exactly the operational moves the qualitative tradition has been negotiating. The novel contribution this paper adds is the connection between the formalization and the alignment programme: the same channels the welfare-economics tradition uses to close measurement gaps are the channels that, under the Goodhart machinery of Section 4, tighten alignment guarantees in the direction of the operational truth. Better economic models are not just better welfare measurement; they are tighter alignment guarantees, and the tightening mechanism is the same one welfare economics has been refining since SSF 2009.

## 6 Empirical Testability via Revealed Sacrifice

The quantification thesis of Section 5 is empirically operational because the [Lasser, 2026d, Revealed Sacrifice] channel maps directly onto observable economic ledgers. Purchase records, time-use surveys, labour-supply panels, and public-commitment ledgers are revealed-sacrifice data in the GFM sense; the [Lasser, 2026e, Need Sufficiency] diagnostic architecture (gap decomposition, wireheading-consistent HHI, need-access cost) is the interpretation layer that turns raw observation into testable predictions. This section catalogues the formal instruments and operationalizes the Goodhart theorem of Section 4 as a cross-domain empirical test. The worked examples of Section 7 then illustrate the instruments concretely.

A note on the relationship to existing operational welfare indices. The Human Development Index [Anand and Sen, 1994], the OECD Better Life Index, the UNDP Multidimensional Poverty Index, and similar deployment-readable welfare instruments are operational measurement tools rather than formal frameworks: they aggregate chosen dimensions through chosen weighting schemes to produce deployment-readable scalars. The instruments below are *formal analogues* to these operational indices and supply an interpretation layer for analogous index choices: deployments using the operational indices can read the indices’ implicit individuation and weighting decisions through

| Precursor                       | Qualitative content                                                                        | GFM quantification                                                                                     | Channel                    |
|---------------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------------------|
| Sen 1985                        | Capability set, functioning vector, capability/functioning gap                             | Poset $P$ , $\text{vol}_P$ , $\text{vol}_R^{[W]}$ , five-cell $\delta$ -decomposition                  | All four                   |
| Nussbaum 2000                   | Basic vs higher capabilities (normative)                                                   | Polarity boundary + downstream-safety condition                                                        | Channel 2                  |
| Becker 1965                     | Household production $F_j$ (unconstrained)                                                 | M6 cooperative caps (poset-coherent) + $\alpha_{n,c}$                                                  | Channel 4                  |
| Bernheim–Rangel 2009            | Choice-domain welfare-revelation distinction (behavioral)                                  | Polarity boundary + S1-attestation as capability-sufficiency split                                     | Channel 2                  |
| SSF 2009                        | GDP-misses-non-market gap (qualitative)                                                    | ResS + five-cell $\delta$ -decomp (deployment-readable)                                                | Channels 2, 3              |
| Alkire–Foster 2011              | Multidimensional poverty index (counting / weighted-sum aggregation)                       | M6 cooperative excess + polarity boundary as additional structure                                      | n/a (already quantitative) |
| Basu–López-Calva 2011           | Survey of capability-set formalizations (cardinality, indirect-utility, effective-freedom) | Poset measure with M6 + polarity as structural upgrade beyond surveyed approaches                      | n/a (formalization survey) |
| Sugden 2003 / Pattanaik–Xu 2000 | Opportunity-set ranking axiomatics; Sugden’s measurement-impossibility                     | GFM measures realized exercise + capability volume, not opportunity-as-individuality                   | n/a (structural neighbor)  |
| Arrow 1951 / Adler 2012         | Ordinal-preference impossibility; prioritarian SWF                                         | $\text{vol}_P$ as measure on poset (appears to sidestep original framing); not a preference aggregator | n/a                        |

Table 2: The capability-economics lineage mapped onto GFM quantifications and the four channels of Sections 5.2–5.5. The SSF 2009 row also flags the diagnostic-decomposition role: the five-cell  $\delta$ -attribution acts as a deployment-readable selector for which channel binds, mirroring the role the SSF decomposition plays in welfare-economics measurement methodology. The Arrow 1951 row is structural rather than channel-bound: GFM appears to sidestep the impossibility by being a measure on a poset rather than a ranking aggregator, but whether an analogous impossibility result applies to measure-aggregation on posets is open (Open Question 8.4).

the GFM apparatus to identify which channel of Section 5 is operative when the index moves. The match between a specific operational index and a specific GFM instrument is not exact in general (the operational indices were not designed to be GFM-compatible), but the structural correspondence is enough to position the operational indices and the formal instruments as complementary rather than competing.

## 6.1 Instruments: GFM machinery as economic-data interpretation

Six formal instruments interpret revealed-sacrifice data. Each maps a familiar economic data source to a specific component of the formalization, and each is what one or more channels of Section 5 act on.

**Two channels (money + time).** [Lasser, 2026d, Section 4] establishes two substantively distinct observation modalities: money sacrifice (priced trade events) and time sacrifice (foregone-wage opportunity costs). The two channels triangulate the same underlying realized exercise; cross-channel agreement is a robustness check, and divergences flag attestation or individuation issues. Predictions 1 through 3 of Section 3 are testable through either channel, with cross-channel triangulation providing independent confirmation.

**Aggregate lower bound  $\beta^{\text{lower}}$ .** [Lasser, 2026d, Theorem 2] bounds the realized exercise volume from below by an aggregate of admissible revealed-sacrifice events.  $\beta^{\text{lower}}$  is the aggregate lower

| Instrument                                   | Economic data source                                                          | What it measures                                                                                                   | Channel(s)                          |
|----------------------------------------------|-------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| Two channels (money + time)                  | Purchase ledgers + time-use surveys                                           | Independent observation modalities of revealed sacrifice                                                           | Channel 1                           |
| Aggregate lower bound $\beta^{\text{lower}}$ | Trade-event ledger restricted to admissible events                            | Diagnostic ceiling $(1 - \beta^{\text{lower}})/\beta^{\text{lower}}$ on $\varepsilon_{\text{gap}}^{\text{nonres}}$ | Channels 1, 2                       |
| Five-cell $\delta$ -decomposition            | Cell-wise attribution across covered, dormant, restricted, residual, boundary | Where the gap lives across structural categories                                                                   | Selector for Channels 1–4           |
| Wireheading-consistent HHI                   | Population-scale trade-flow concentration                                     | Concentration consistent with Conjecture 1                                                                         | Diagnostic for Channel 1 saturation |
| Need-access cost (NAC)                       | Below-sufficiency expenditure ledger                                          | Need-side share of revealed sacrifice                                                                              | Channels 3, 4                       |
| Per-event $\alpha_{n,c}$                     | Bundle decomposition extension                                                | Per-capability attribution within bundles                                                                          | Channel 4                           |

Table 3: The six formal instruments connecting GFM machinery to economic data sources. Each instrument operationalizes one or more components of  $\varepsilon_{\text{gap}}$  and is what at least one channel of Section 5 acts on. The five-cell  $\delta$ -decomposition plays a selector role across all four channels, mirroring the diagnostic role the SSF 2009 decomposition plays in welfare-economics measurement methodology.

ratio;  $(1 - \beta^{\text{lower}})/\beta^{\text{lower}}$  is the corresponding deployment-readable diagnostic ceiling on the within-channel disagreement gap  $\varepsilon_{\text{gap}}^{\text{nonres}}$  (Remark 17). Computing  $\beta^{\text{lower}}$  requires: (i) a defined trade window  $W$ , (ii) classification of events as admissible or not under S0, S1, S4(a), and the genuine-exchange conditions, and (iii) aggregation under Part A (bundle-level) or Part B (per-capability) modes.

**Five-cell  $\delta$ -decomposition.** [Lasser, 2026e, Definition 8] attributes the gap across  $\delta_{\text{covered}}$  (channel observed),  $\delta_{\text{dormant}}$  (channel could reach but hasn’t),  $\delta_{\text{restricted}}$  (deliberately non-exercised),  $\delta_{\text{residual}}$  (ResS proper), and  $\delta_{\text{boundary}}$  (boundary-residual). The attribution is a structure question (“where does the gap live?”) and is the deployment instrument that selects which channel of Section 5 binds for the given gap profile.

**Wireheading-consistent HHI.** [Lasser, 2026e, Propositions 6 and 7] construct a Schur-convex concentration measure on trade-flow distributions across bundle categories. The HHI is the population-scale empirical witness for Conjecture 1: high HHI indicates concentration consistent with optimization-pressure exploitation of  $P$ - $T$  divergence regions, though as Remark 23 insists, the witness is correlational rather than causal.

**Need-access cost (NAC).** [Lasser, 2026e, Definition 9] tracks below-sufficiency expenditure: revealed sacrifice on capabilities below the polarity boundary, where retention is degrading and the agent has no non-degrading alternative. NAC distinguishes need-share from want-share contributions to revealed sacrifice and is the operational substrate for Prediction 5.

**Per-event  $\alpha_{n,c}$ .** [Lasser, 2026d, Definition 4] defines the event-local bundle decomposition coefficients  $\alpha_{n,c}$  that partition a bundle’s contribution to  $\Delta \text{vol}_P(X_n)$  across capability components. Per-event decomposition is the substrate for Channel 4 and for Predictions 1 and 2.

## 6.2 Operationalizing the Goodhart bound

The Goodhart theorem of Section 4 is testable cross-domain and longitudinally. Both tests follow the same structural recipe but ask different questions of the data.

**Cross-domain test.** For a collection of capability domains  $\mathcal{D}_1, \dots, \mathcal{D}_m$  in which the framework operates, compute the diagnostic ceiling  $(1 - \beta^{\text{lower}})/\beta^{\text{lower}}$  on each domain's  $\varepsilon_{\text{gap}}^{\text{nonres}}$  and the cell-wise attribution via the  $\delta$ -decomposition. For a fixed alignment property  $g$  (say, anti-monopolar threshold accuracy), measure  $g$ 's deviation from a domain-specific operational truth. Theorem 2 certifies that domains with smaller diagnostic-ceiling values support smaller *certified upper bounds* on  $|g(T) - g(P)|$ ; the cross-domain empirical test is therefore a directional-consistency check between the diagnostic ceiling and observed deviations, with the cell-wise attribution telling the deployer which channel intervention would produce the within-domain tightening of the certified bound. This test does not require closing the per-property Lipschitz constants; it requires only checking the directional consistency between cross-domain  $\varepsilon_{\text{gap}}^{\text{nonres}}$  and cross-domain  $|g(T) - g(P)|$ . A domain violating the directional consistency is either a Lipschitz failure (the property's sensitivity to  $\varepsilon_{\text{gap}}^{\text{nonres}}$  is non-Lipschitz on the domain) or an attestation failure (the  $\beta^{\text{lower}}$  ceiling is unreliable for the domain).

**Longitudinal test.** For a single deployment over a window  $W = [t_0, t_0 + T]$  in which the deployment performs an admissible intervention on one of the four channels (more events, sharper attestation, re-individuation, bundle decomposition), measure the intervention's effect on  $\beta^{\text{lower}}$ , on the cell-wise attribution, and on  $g$ 's deviation from the operational truth. Theorem 2 certifies a monotone tightening of the *upper bound* on  $|g(T) - g(P)|$  with the intervention's effect on  $\varepsilon_{\text{gap}}^{\text{nonres}}$ ; the empirical test is the directional-consistency check on observed deviations. Failure of monotonicity in the certified bound is either a Lipschitz violation, an attestation regression, or an unaccounted-for counter-channel effect (e.g., re-individuation that increases within-channel disagreement at the same time as it decreases the floor; cf. Remark 27).

**Distinguishing test outcomes.** Three distinct test outcomes have distinct interpretations:

1. *Tightening as predicted, attribution consistent with intervention.* The framework's prediction is supported, and the deployment can plan further interventions on the binding channel.
2. *Tightening as predicted, attribution inconsistent with intervention.* The intervention worked, but for different reasons than expected. This is informative: it suggests the cell-wise attribution is mis-specified (the intervention reduced the wrong cell's gap), which is a diagnostic-decomposition issue rather than a Goodhart issue.
3. *No tightening or anti-tightening.* The intervention failed. The most common failure modes are: (a) Lipschitz violation (the alignment property is not Lipschitz on the relevant capability subspace under the deployment's individuation), (b) adversarial attestation degradation (Open Question 8 of Section 8), and (c) cross-channel offset (one channel improved while another worsened). Distinguishing the failure modes requires cell-wise attribution before and after the intervention.

**Sup-norm coarseness and empirical sensitivity.** A caveat for the empirical programme that does not affect the theory. The sup-norm definition of  $\varepsilon_{\text{gap}}^{\text{nonres}}$  (Definition 3) is conservative (Remark 16): a deployment that tightens  $\varepsilon_{\text{gap}}^{\text{nonres}}$  on 99 % of capability dimensions while slightly worsening it on one shows no improvement under the sup-norm. How this caveat propagates to the diagnostic ceiling  $(1 - \beta^{\text{lower}})/\beta^{\text{lower}}$  depends on how the deployment reads the ceiling. When the deployment allocates per-capability ceiling components under Assumption 1 part (i) and then takes the worst-case across them as a sup-norm proxy, the diagnostic ceiling inherits the worst-dimension conservatism; when the deployment reads the ceiling purely as an aggregate (without per-capability allocation), the aggregate may still register average progress even if a single dimension regresses, but the bound on the sup-norm  $\varepsilon_{\text{gap}}^{\text{nonres}}$  is then no longer tight (Remark 17). In practice the cross-domain and longitudinal tests above may therefore be *insensitive to genuine progress on most dimensions when a single dimension is regressing* (under the sup-norm reading that bounds  $\varepsilon_{\text{gap}}^{\text{nonres}}$  tightly), and a deployment may be meaningfully improving without the sup-norm-bounded diagnostic registering it. This is a feature for safety-conservative alignment monitoring (one badly-proxied dimension is sufficient to produce a misaligned decision), but it is a limitation for deployment-progress measurement, where weighted norms or per-dimension diagnostics may be operationally more informative. Open Question 8.4 flags the weighted-norm direction as a candidate refinement; the empirical operator should interpret a flat sup-norm-bounded diagnostic under intervention as “no progress on the worst dimension” rather than as “no progress at all,” and may choose to track the aggregate  $\beta^{\text{lower}}$  separately as a deployment-progress signal that the sup-norm bound's conservatism does not eat.

### 6.3 Testing the four channels

Each of the four channels of Section 5 has a specific test that the instruments support.

**Channel 1 (observation density).** The longitudinal trajectory of  $\beta^{\text{lower}}$  under admissible accumulation should be non-decreasing ([Lasser, 2026d, Propositions 2 and 3]). Concretely: add a sequence of admissible trade events to a deployment’s ledger one at a time;  $\beta^{\text{lower}}$  at each step should be at least as large as the previous step. A decrease falsifies either the admissibility classification or the framework’s non-decreasing claim. In practice, deployment data is added in batches; batch-wise non-decreasing is the operational version of the test.

**Channel 2 (attestation quality).** The attested-share of admissible events (events that pass the S0, S1, S4(a) checks under the deployment’s institutional attestation infrastructure) should rise as the attestation layer matures. The diagnostic-ceiling  $(1 - \beta^{\text{lower}})/\beta^{\text{lower}}$  at fixed observation count should tighten as the attested share rises (events previously classified as non-admissible become admissible under sharper attestation). Falsification:  $\beta^{\text{lower}}$  does not respond to attested-share improvements, indicating that the attestation layer’s classification is not improving the operational admissibility set or that the events being attested are non-binding.

**Channel 3 (individuation discipline).** A re-individuation move (e.g., treating household labour as a distinct capability with imputed-time pricing) should reduce  $\varepsilon_{\text{floor}}^{\text{res}}$  and add a corresponding contribution to  $\varepsilon_{\text{gap}}^{\text{nonres}}$  from the newly-admitted capability. The net effect on  $\varepsilon_{\text{gap}} = \varepsilon_{\text{gap}}^{\text{nonres}} + \lambda \cdot \varepsilon_{\text{floor}}^{\text{res}}$  depends on whether the new capability’s attested share is high (in which case its  $\varepsilon_{\text{gap}}^{\text{nonres}}$  contribution is small, and the net move is a tightening) or low (in which case the move is neutral or even alignment-loosening, with the  $\lambda$ -weighted net  $\Delta_3^\lambda$  approaching zero or going negative; cf. Section 8.3). Test: pre/post comparison of  $\varepsilon_{\text{floor}}^{\text{res}}$ ,  $\varepsilon_{\text{gap}}^{\text{nonres}}$ , and the alignment property; the move tightens iff the new capability’s attested share exceeds a deployment-specific threshold determined by  $\lambda$ .

**Channel 4 (bundle decomposition).** The per-event  $\alpha_{n,c}$  decomposition (when the deployment’s commitment scheme supports it) should produce within-bundle component-wise lower bounds that, when aggregated, are at least as tight as the bundle-level Part-A aggregation of [Lasser, 2026d, Theorem 2]. Operationally: compute  $\beta^{\text{lower}}$  under both Part-A and Part-B modes for the same trade-event set; Part-B (per-capability) should be at least as informative as Part-A on the within-bundle poset-independence subset of events. If Part-B is tighter, Channel 4 is delivering its predicted tightening; if not, the bundle-decomposition extension is either inactive or applied to events outside its admissibility scope.

### 6.4 Methodology template and forward

[Lasser, 2026e, Section 6] works through a four-phase end-to-end demonstration of the [Lasser, 2026e, Need Sufficiency] diagnostic architecture on a single domain: capability inventory  $\rightarrow$  trade-event ledger  $\rightarrow$  cell-wise attribution  $\rightarrow$  deployment intervention. That worked example is a methodology template. The four worked examples of Section 7 below illustrate the full machinery of the present paper at four scales: (i) micro-level divergence between a GFM prediction and the utility-theoretic account (Section 7.1), (ii) the Goodhart gradient under variable observation infrastructure (Section 7.2), (iii) fungibility-collapse limit at macro scale (Section 7.3), and (iv) mixed-polarity bundle-for-bundle trades with deployment misalignment consequences (Section 7.4). Each example exercises specific instruments from Table 3 and demonstrates concrete tightening predictions or failure modes; the four examples together cover all six instruments and all four channels.

## 7 Worked Examples

The four worked examples below illustrate the formalization at different scales and exercise different combinations of the instruments from Section 6. Each example follows the same structure: scenario setup, the standard utility-theory account (where applicable), the GFM account computed through the relevant instruments, the operative channels, and either a concrete testable prediction or a deployment-misalignment consequence.

## 7.1 The boat purchase: micro-level divergence (Channel 4)

**Scenario.** A rational agent with discretionary purchasing power chooses to spend a substantial fraction on a depreciating asset: a small recreational boat. The choice puzzles a standard-utility account because (i) the boat depreciates rapidly, (ii) it has limited resale liquidity, and (iii) the agent has alternative investments with higher expected monetary returns. Why does the agent buy the boat?

**Utility-theory account.** Under standard utility theory, the trade is rationalizable by asserting  $u_{\text{agent}}(\text{boat}) > u_{\text{agent}}(\text{cash})$  at the price paid. The assertion is a post-hoc utility assignment, not a structural prediction: the same agent’s utility could be assigned any shape consistent with the observed choice, and standard utility theory does not constrain  $u(\cdot)$  from the agent’s observable capability inventory. Two agents who appear identical on every observable demographic dimension can have arbitrarily different  $u(\text{boat})$ , and the framework offers no way to predict which agent would buy.

**GFM account.** A boat is not a self-contained capability; it is a multiplier that operates on a set of prerequisite capabilities the agent already holds. The bundle the boat *activates* is something like {water-access (prerequisite, not provided by the boat), recreational outdoor activity, peer activities with boat-owners (cooperative capability), waterfront leisure time}, and the boat’s contribution is structurally tied to whether these prerequisites and complements are present in the agent’s pre-trade poset  $P_i$ .

Under M6 (cooperative capabilities, [Lasser, 2026g, Proposition 1]), the cooperative excess from adding boat ownership to a poset is *high* when complementary capabilities are already present and *low* when they are absent. The same boat at the same priced cost produces different per-event bundle-decomposition coefficient profiles [Lasser, 2026d, Definition 4] across agents: an agent with established water-access (lakeside residence) and an existing boat-owner social network realizes high cooperative gain, because the boat completes a combination its prerequisites already make valuable; an agent in a city without nearby water access and no boating network realizes low cooperative gain, because the boat’s components cannot be activated from the agent’s existing poset and the boat sits in the poset as an isolated capability with limited reach.

The structural prediction is that cross-agent variance in revealed-sacrifice magnitudes for boat purchases is generated by poset-context dependence rather than ad hoc utility shape: agents with strong complementary pre-trade capabilities pay *more* than agents without, by an amount that scales with the cooperative-gain difference  $\Delta \text{vol}_{\text{P}_{\text{coop}}}(\text{boat} \mid P_{\text{rich}}) - \Delta \text{vol}_{\text{P}_{\text{coop}}}(\text{boat} \mid P_{\text{sparse}})$ , where  $P_{\text{rich}}$  is the complement-rich poset and  $P_{\text{sparse}}$  the complement-sparse one. This is the cross-agent variance signature of Prediction 2 (cooperative premium) realized in a single bundle: the variance is high because the cooperative components depend on poset context.

The complementary direction of cooperative-capability dependence (complements amplify the marginal contribution) is structurally distinct from the substitutive direction of Prediction 1 (substitutes reduce the marginal contribution), but both come from the same M6 mechanism. The boat is a complement-leveraging case; a closet of redundant winter coats would be a substitution case. Both sides are GFM-native predictions that utility theory cannot generate without ad hoc utility-shape assumptions.

### Operative channels and instruments.

- **Channel 4** (bundle decomposition): the per-component  $\alpha_{n,c}$  coefficients carry the cross-agent variance.
- **Channel 2** (attestation): the test requires consented-perimeter exposure of the agents’ pre-trade capability inventories, which is institutional infrastructure beyond the public-ledger default.
- **Instrument: per-event  $\alpha_{n,c}$ .** The bundle-decomposition extension of [Lasser, 2026d, Open Question 2] is the operational substrate.

**Test.** Cross-sectional revealed-sacrifice magnitudes for boat purchases, regressed against pre-trade complementary-capability indices (proximity to water, social-network density of boat-owners in

the agent’s network, existing outdoor-recreation infrastructure). Standard utility theory predicts no systematic relation; GFM predicts the regression coefficient is *positive* (more complements  $\rightarrow$  higher sacrifice) and structurally constrained by the cooperative-coherence of the agent’s poset. A null result falsifies the prediction; a positive relation whose magnitude exceeds the cooperative-coherence bound suggests an unmodeled complementary capability or poset mis-specification.

## 7.2 The Goodhart gradient: alignment-tightening under observation (Channels 1 and 2)

**Scenario.** Two AI capability-restriction deployments operate in nominally the same domain (a frontier model deployed for a high-stakes application, with restriction landscape governing what capabilities the model is permitted to exercise). The deployments differ in their observation infrastructure:

- **Scenario A.** The framework measures capabilities via a coarse binary proxy (can / cannot exercise capability), with sparse trade observation (few admissible events per capability category) and weak S0 attestation (a substantial share of events are non-admissible due to attestation failures).
- **Scenario B.** The framework measures capabilities via a fine-grained graded benchmark, with dense trade observation across the capability inventory and sharp S0 attestation (institutional infrastructure delivers high attested-share).

Both deployments operate the same alignment-restriction logic on top of the framework; the question is how much the restriction landscape diverges from the operational truth in each.

**Computation through the instruments.** In Scenario A, the diagnostic ceiling on  $\varepsilon_{\text{gap}}^{\text{nonres}}$  is  $(1 - \beta_A^{\text{lower}})/\beta_A^{\text{lower}}$ , with  $\beta_A^{\text{lower}}$  low (illustratively,  $\beta_A^{\text{lower}} \approx 0.4$ , giving an  $\varepsilon_{\text{gap}}^{\text{nonres}}$  ceiling of 1.5). The  $\delta$ -decomposition shows substantial mass in  $\delta_{\text{dormant}}$  (capabilities the channel could reach but hasn’t observed),  $\delta_{\text{restricted}}$  (capabilities deliberately non-exercised under risk-management), and a wide  $\delta_{\text{covered}}$  band that includes both proxy-failure-candidates and observation slack.  $\varepsilon_{\text{floor}}^{\text{res}}$  is moderate. In Scenario B,  $\beta_B^{\text{lower}}$  is high ( $\beta_B^{\text{lower}} \approx 0.85$ ,  $\varepsilon_{\text{gap}}^{\text{nonres}}$  ceiling  $\approx 0.18$ );  $\delta_{\text{covered}}$  is small,  $\delta_{\text{dormant}}$  is small (channel reach is mature), and  $\delta_{\text{residual}}$  is at the deployment’s structural floor.

By Theorem 2, each Lipschitz alignment property’s deviation from the operational truth is bounded by  $\text{Lip}(g) \cdot \varepsilon_{\text{gap}}^{\text{nonres}}$ , so Scenario B’s bound is roughly  $1.5/0.18 \approx 8\times$  tighter than Scenario A’s at the diagnostic-ceiling level (the multiplier is application-dependent through  $\text{Lip}(g)$ ; the headline is the order-of-magnitude difference rather than any specific factor). Concretely: the anti-monopolar threshold  $\gamma^*$  is estimated tightly in B, the phase boundary is accurately located, and the restriction landscape tracks the operational-truth landscape closely; in A, the threshold has wide error bars, the phase boundary is conservatively placed (over-cautious in the [Lasser, 2026i, Wamura] pattern), and the restriction landscape is systematically over-restrictive.

### Operative channels and instruments.

- **Channel 1** (observation density): drives  $\delta_{\text{dormant}}$  from large (A) to small (B).
- **Channel 2** (attestation quality): drives the attestation-limited share of  $\delta_{\text{covered}}$  from large (A) to small (B).
- **Instruments:** aggregate  $\beta^{\text{lower}}$ , five-cell  $\delta$ -decomposition, alignment-property estimation uncertainty.

**Test and tightening attribution.** The cross-domain Goodhart test of Section 6.2 predicts that B’s alignment guarantees are tighter than A’s by  $\text{Lip}(g)$  times the difference in  $\varepsilon_{\text{gap}}^{\text{nonres}}$  ceilings between the two scenarios on each Lipschitz property  $g$ . The cell-wise attribution tells the deployer *which* infrastructure investment produced the tightening. In a transition from A to B, attribute the gap reduction across:  $\delta_{\text{dormant}}$  reduction (Channel 1 working), attestation-limited  $\delta_{\text{covered}}$  reduction (Channel 2 working), and any residual  $\delta_{\text{covered}}$  proxy-failure-candidate reduction (mixed Channel 1/2 effect or genuine proxy improvement). The attribution is what turns a qualitative claim (“B is better”) into a structurally precise one (“B has 70% less  $\delta_{\text{dormant}}$  and 60% less attestation-limited  $\delta_{\text{covered}}$ , so the tightening is mostly Channel 1 with a meaningful Channel 2 contribution”).

### 7.3 The fungibility collapse: standard economics as macro limit (no channel)

**Scenario.** A representative-agent macroeconomic application: aggregate consumer welfare, GDP-weighted, on a population of agents trading approximately fungible goods through liquid markets at representative-agent scales of aggregation. The standard money-maximization model has been empirically successful for this regime: consumption smoothing under liquidity constraints, capital allocation across sectors, and intertemporal substitution at aggregate level all fit reasonably well. Why?

**Computation through the instruments.** The fungibility-collapse correspondence (Theorem 1) explains the phenomenon. At macro scale on  $P_{\text{std}}$ , the cooperative term in M6 vanishes, individuation collapses to representative agents, and the poset degenerates to the wealth axis. GFM’s  $\text{vol}_P$  on  $P_{\text{std}}$  reduces to total wealth, and  $\text{vol}_P$ -maximization reduces to expected-wealth maximization. Consequently, on the observed-and-non-residual subset, the proxy-truth gap  $\varepsilon_{\text{gap}}^{\text{nonres}}$  approaches zero: the proxy is effectively exact when the poset is fungible, because the five structural features GFM adds beyond  $P_{\text{std}}$  (non-fungibility, cooperative capabilities, anti-monopolar reasoning, individuation, residual recognition) all become operationally irrelevant.

What does *not* go to zero under collapse:  $\varepsilon_{\text{floor}}^{\text{res}}$ . The residual class ResS remains non-empty. Household production, volunteer activity, gift-economy capabilities, informal economy activity, dark-market activity all remain outside the channel’s reach even when the observed subset’s poset is fully fungible. Standard economics knows this; SSF 2009 documents the gap at length [Stiglitz et al., 2009]. The gap is a channel-reach limit, not a poset-structure limit, and the fungibility-collapse correspondence does not extend the channel’s reach.

#### Operative channels and instruments.

- **No within-channel channels operative.** The collapse happens to be the regime where Channels 1, 2, 4 have nothing to tighten because the observed-and-non-residual subset’s  $\varepsilon_{\text{gap}}^{\text{nonres}}$  is already approximately zero.
- **Channel 3** (individuation) is the only channel with content: re-individuating capabilities currently in ResS (treating household production as a distinct imputed-time-priced capability, etc.) reduces  $\varepsilon_{\text{floor}}^{\text{res}}$  by  $\Delta_3^{\text{floor}}$  at the cost of adding  $\Delta_3^{\text{new}}$  to the within-channel ceiling, with net effect  $\Delta_3^\lambda = \lambda \cdot \Delta_3^{\text{floor}} - \Delta_3^{\text{new}}$  on the alignment slack. Channel 3 tightens iff  $\Delta_3^\lambda \geq 0$ , i.e., when the imputation’s measurement slack does not exceed  $\lambda$  times the floor reduction. This is the welfare-economics imputation programme [Stiglitz et al., 2009, Aguiar and Hurst, 2007] read through the GFM lens.
- **Instruments:**  $\delta_{\text{residual}}$  (cell mass at the floor),  $\varepsilon_{\text{floor}}^{\text{res}}$  (deployment-specific share).

**Lesson.** Standard economics’ empirical success at macro scale is exactly the regime where (i) the observed poset is approximately fungible, (ii) the residual class is small enough to neglect under representative-agent aggregation, and (iii) cooperative / individuation structure is operationally insignificant. When any of the three conditions fails (micro-scale bundle decisions, non-market sectors, cross-cultural welfare comparison, multi-substrate AI capability landscapes), the standard model breaks down and the five structural features become load-bearing (Section 2.4). The fungibility-collapse correspondence makes the breakdown structurally visible: it tells the reader where the standard model is right (and why) and where it fails (and through which feature).

The example also illustrates that “better economic models  $\rightarrow$  tighter alignment” is not a uniform claim across deployment regimes. At macro scale on  $P_{\text{std}}$ , alignment is already tight by Theorem 1 on the observed subset and bounded below by  $\lambda \cdot \varepsilon_{\text{floor}}^{\text{res}}$  on the residual; further within-channel tightening has no content because the within-channel gap is already approximately zero. The binding constraint is the floor, and the operational lever is Channel 3 (re-individuation of non-market capabilities). This is the same operational lever welfare economics identified in 2009.

### 7.4 Residential relocation: mixed-polarity bundle-for-bundle trades (Channels 3 and 4)

**Scenario.** An agent moves from home A to home B at the same nominal purchase price. The trade surrenders bundle  $X = \{\text{shelter}_A, \text{neighborhood}_A, \text{commute}_A, \text{school}_A, \dots\}$  and acquires



bundle  $Y = \{\text{shelter}_B, \text{neighborhood}_B, \text{commute}_B, \text{school}_B, \dots\}$ . Two agents perform nominally identical priced relocation trades. Agent  $A$ 's pre-trade state has shelter at sub-sufficiency (foreclosure-driven relocation, with retention degrading sufficiency and no non-degrading third option available); agent  $J$ 's pre-trade state has shelter at above-sufficiency (job-driven relocation, with retention non-degrading and multiple non-degrading alternatives available).

**Utility-theory account.**  $\Delta U = U(B) - U(A) > 0$  for both agents (each chose B). No further structural decomposition. Two agents performing the same priced trade are treated identically under linear utility; under heterogeneous concave  $u_i$ , the variance is attributed to  $u_i$ 's shape rather than to the agents' pre-trade need-satisfaction states. The framework cannot generate the bifurcation prediction.

**Computation through the instruments.** By [Lasser, 2026e, Definition 3], the polarity boundary on the shelter dimension distinguishes need-side (below sufficiency, retention degrading) from want-side (above sufficiency, retention non-degrading). Agent  $A$ 's shelter trade is below the polarity boundary on shelter; the sacrifice magnitude on the shelter component reveals cost-of-access, captured by NAC ([Lasser, 2026e, Definition 9]). Agent  $J$ 's shelter trade is above the polarity boundary; the sacrifice magnitude reveals preference, captured by the standard revealed-sacrifice channel. The non-shelter components (neighborhood, commute, school) are above sufficiency for both agents and contribute want-side sacrifice.

The bundle decomposition gives the per-component split. Agent  $A$ 's split has high need-share:  $\text{NAC}_A(Y)/\text{Sacrifice}_A(X) \gg \text{NAC}_J(Y)/\text{Sacrifice}_J(X)$ . The S1-attestation layer's H1 (retention degrading) and H2 (no non-degrading alternative) classifications place agent  $A$  in the below-sufficiency-with-no-alternative regime ([Lasser, 2026e, Definition 6]); agent  $J$  remains in the above-sufficiency regime. The classification is what bifurcates the population into the two regimes Prediction 5 predicts.

#### Operative channels and instruments.

- **Channel 3** (individuation): distinguishing need-share from want-share on the shelter dimension requires individuating “shelter at or below sufficiency” as a capability with NAC accounting separate from “shelter above sufficiency.” The polarity boundary is the individuation cut.
- **Channel 4** (bundle decomposition): the per-component  $\alpha_{n,c}$  decomposition is what enables the shelter-vs-non-shelter split. Without it, the trade is observed as a single priced bundle and the bifurcation is invisible.
- **Instruments:** NAC, S1-attestation, polarity boundary, per-event  $\alpha_{n,c}$ .

**What the bifurcation enables.** The framework's structural ability to distinguish agent  $A$  from agent  $J$  on nominally identical priced trades is operationally significant in several ways, none of which require taking a normative position on which agent should be favored:

1. *Distributional measurement.* The framework yields a population-level decomposition of relocation trades by pre-trade need-satisfaction state. Utility theory cannot extract this signal from priced-trade data alone: it sees only the price and the chosen direction. GFM sees the fraction of aggregate sacrifice that is need-driven (NAC) versus want-driven (above-sufficiency). This distribution is a measurement artifact in its own right, prior to any policy conclusion drawn from it.
2. *Disparate-incidence diagnosis.* Any policy that operates on a single proxy of relocation activity (relocation tax, mobility subsidy, displacement compensation) treats both regimes identically. The bifurcation makes the differential incidence of the uniform-proxy policy structurally visible: the policy's effect on the high-NAC-share cluster differs from its effect on the low-NAC-share cluster, by an amount the framework can quantify. Whether the differential is unjust, intended, or acceptable is a normative question the framework does not answer; whether it exists is now a question with a formal answer.
3. *Structural needs-test.* The H1+H2 classification (retention degrading + no non-degrading alternative) is a structural needs-test that operates on the trade itself rather than on ad hoc

demographic eligibility cutoffs. Eligibility for need-targeted assistance, if a deployment chooses to target need, follows from the structure of each trade rather than from income thresholds, age brackets, or household-composition rules. This does not say the framework’s needs-test is the right one to use; it says the framework provides a needs-test that is structural rather than categorical, and a deployment can compare it against whatever categorical test it currently uses.

4. *Welfare-aggregation hygiene.* Aggregate welfare measures pooling need-driven and want-driven trades inherit a polarity-mixed signal in which sacrifice on the high-NAC side reveals cost-of-access while sacrifice on the low-NAC side reveals preference. The bifurcation tells the analyst what fraction of the aggregate is preference-revealing versus cost-of-access, which is information the analyst needs to interpret the aggregate honestly.

**Tests for the bifurcation.** Three concrete tests of the bifurcation hypothesis itself:

1. *Bifurcation test.* Cross-sectional analysis of agents performing similar priced relocations, stratified by pre-trade need-attestation. Prediction: NAC-share versus above-sufficiency-share splits cluster bimodally on shelter sufficiency rather than continuously on the priced trade magnitude.
2. *S1-attestation test.* H1+H2 classification should partition the agent population on shelter sufficiency consistently with the bifurcation in test 1; agents whose H1+H2 status indicates degrading retention with no non-degrading alternative should be the high-NAC-share cluster.
3. *Population HHI test.* The wireheading-consistent HHI on relocation sub-categories should distinguish forced-relocation regions (high HHI, mass concentrated in foreclosure / displacement categories) from voluntary-mobility regions (lower HHI, mass distributed across job-driven, family-driven, lifestyle-driven categories). This is Prediction 4 applied to the relocation sub-domain.

**Goodhart deployment misalignment.** A deployment that optimizes a coarser proxy than the formalization warrants (e.g., a relocation-tax framework optimizing “average price paid for relocation”, a vol<sub>P</sub> proxy with no need-share / want-share distinction) treats both regimes identically. Under Theorem 2, the proxy-truth gap on this deployment includes the entire below-sufficiency contribution to NAC, and the misalignment magnitude is bounded by  $\text{Lip}(g) \cdot \epsilon_{\text{gap}}^{\text{nonres}}$  on the relocation-tax outcome. Sharper economics (admitting bundle decomposition and polarity boundary into the framework) delivers a tighter alignment to the true distribution of welfare consequences. We emphasize: this does not say the deployment *should* treat the two regimes differently; it says that if the deployment treats them identically, the cost in alignment-target accuracy is bounded by Theorem 2 and the bound is not zero.

A reading caveat on the bound. The misalignment magnitude  $\text{Lip}(g) \cdot \epsilon_{\text{gap}}^{\text{nonres}}$  is a function of the deployment’s chosen individuation: a deployment that has *already* performed the shelter-at-vs-above-sufficiency individuation (drawn the polarity boundary) has a smaller  $\epsilon_{\text{gap}}^{\text{nonres}}$  that distinguishes the two regimes; a deployment that has *not yet* performed the individuation has a larger  $\epsilon_{\text{gap}}^{\text{nonres}}$  on a coarser capability space that does not see the bifurcation at all, with a correspondingly larger but structurally different bound. The example partly motivates the individuation move: the bound’s bite depends on the individuation having occurred, but the comparison between “with bifurcation” and “without bifurcation” bounds is what shows the deployer the cost of staying on the coarser individuation.

This is the canonical case where Channels 3 and 4 jointly deliver alignment-tightening that no single channel could deliver. Channel 3 alone (individuating shelter sufficiency) would not separate the trades’ shelter contributions from their non-shelter contributions; Channel 4 alone (bundle decomposition without polarity discipline) would split the trade into components but not classify the shelter component’s polarity. Per the sequential-intervention regime of Remark 32, we treat the example as the deployment first fixing the individuation discipline that distinguishes shelter sufficiency (Channel 3, contributing its  $\lambda$ -weighted net  $\Delta_3^\lambda$ ), then performing bundle decomposition on the resulting individuation (Channel 4, contributing  $\Delta_4$ ). Under the sequential regime, Proposition 1 applies directly; co-evolution between the two channels would require the correction terms flagged in Remark 32.

## 7.5 What the four examples demonstrate

The four examples cover four scales (micro, AI alignment, macro, bundle-for-bundle) and exercise four combinations of the channels:

| Example                      | Scale             | Operative channels                                   | Headline prediction                                                                                    |
|------------------------------|-------------------|------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| §7.1: Boat                   | Micro             | Ch. 4 (bundle decomp) + Ch. 2 (attestation)          | Complements amplify cooperative excess; cross-agent variance                                           |
| §7.2: Goodhart gradient      | AI alignment      | Ch. 1 (observation density) + Ch. 2 (attestation)    | $\sim 8\times$ ceiling tightening from A to B                                                          |
| §7.3: Fungibility collapse   | Macro             | None within-channel; Ch. 3 (individuation) for floor | $\epsilon_{\text{gap}}^{\text{nonres}} \rightarrow 0$ but $\epsilon_{\text{floor}}^{\text{res}}$ binds |
| §7.4: Residential relocation | Bundle-for-bundle | Ch. 3 (individuation) + Ch. 4 (bundle decomp)        | Bifurcation by pre-trade sufficiency                                                                   |

Table 4: The four worked examples and their channel composition. The examples are deliberately chosen to span the structural variety: a single-event prediction (boat), a cross-domain infrastructure comparison (Goodhart gradient), a regime-limit case (fungibility collapse), and a deployment-misalignment case (residential relocation). Together they exercise all six instruments of Table 3 and demonstrate that the channel composition of Proposition 1 is operationally visible in concrete data structures.

The examples support three reading-level claims about the formalization. First, the channels of Section 5 are operationally distinct: each example’s binding channels are different combinations, and the examples exercise the full channel space across the four cases. Second, the gap decomposition of [Lasser, 2026e, Need Sufficiency] plays its predicted selector role: in each example, the cell-wise attribution tells the deployer which channel intervention applies. Third, the Goodhart machinery of Section 4 is testable in practice: Section 7.2’s cross-domain comparison is the canonical structural test, and Section 7.4’s deployment-misalignment is the canonical concrete consequence.

The examples do not constitute empirical validation. They are *methodology demonstrations*: they show how the formalization applies, what it predicts, and what would falsify it. Empirical validation of the predictions across actual deployments is deferred work; the contribution at this stage is to identify the predictions clearly enough that they can be tested and to establish the methodology template through which the testing operates.

## 8 Discussion and Open Questions

### 8.1 What this paper establishes

- Quantification of the capability welfare economics tradition.** The Sen 1985 capability set, the Sen functioning vector, the Becker household production function, and the SSF 2009 gap-decomposition catalogue all admit measure-theoretic counterparts in the GFM apparatus (poset,  $\text{vol}_P$ ,  $\text{vol}_R^{[W]}$ , M6 cooperative capabilities, ResS, the five-cell  $\delta$ -decomposition). The Alkire–Foster [Alkire and Foster, 2011] multidimensional- poverty methodology (the closest existing quantitative apparatus) is contained as a special case under additional structural commitments (cooperative excess via M6, polarity discipline). Adjacent formal-welfare-economics work (Bernheim and Rangel [2009] on behavioral welfare economics, Basu and López-Calva [2011] on prior capability-set formalizations, Sugden [2003] and Pattanaik and Xu [2000] on opportunity-set axiomatics, Adler [2012] on prioritarian SWF) is engaged in Section 5.7 as structural neighbors rather than as channel-bound precursors. The positioning is analogous to the Bentham-to-expected-utility formalization that the welfare-economics literature spent roughly a century-and-a-half completing for utility theory.
- GFM strictly contains standard utility theory at the linear / risk-neutral case (containment result)** (Theorem 1). Under fungibility collapse (the structural reduction in which the

capability poset degenerates to a single purchasing-power dimension, cooperative capabilities vanish, and individuation becomes irrelevant), an actor maximizing  $\text{vol}_P$  on  $P_{\text{std}}$  is observationally equivalent to a linear-utility expected-utility maximizer. Together with the five-features catalog of Section 2.4, this establishes containment (utility theory  $\subset$  GFM at  $P_{\text{std}}$ ) plus extension (GFM has structural content beyond  $P_{\text{std}}$  that utility theory lacks). Concavity, risk-aversion, and heterogeneous utility recover through additional calibrations.

3. **Falsifiable structural-prediction class** (Section 3). Five predictions (bundle-completion magnitude, cooperative premium, saturation transition, population-scale anti-monopolar, mixed-polarity bundle-for-bundle) that utility theory either does not make, or can match only by adding ad hoc unfalsifiable structure. The predictions are what becomes falsifiable when the qualitative tradition is quantified.
4. **Goodhart theorem for proxy-instrumental alignment** (Theorem 2). Each Lipschitz alignment property in the GFM sequence, viewed as a function of the proxy  $\text{vol}_P$ , deviates from its value under the operational truth  $T = \text{vol}_R^{[W]}$  (under [Lasser, 2026d, Revealed Sacrifice]’s S0) by at most  $\text{Lip}(g) \cdot \varepsilon_{\text{gap}}^{\text{nonres}}$ . Optimization pressure correlates with gap-exploitation as a qualitative direction; the wireheading-consistent HHI of [Lasser, 2026e, Need Sufficiency] is the deployment-readable empirical witness.
5. **Better economic models tighten alignment, in four operationally distinct channels** (Corollary 1, Proposition 1). Conditional on per-capability admissibility (Assumption 1), the four channels (observation density (Channel 1), attestation quality (Channel 2), individuation discipline (Channel 3), and bundle decomposition (Channel 4)) each deliver a reduction to a non-overlapping component of the alignment-slack diagnostic. Channel 3 contributes through the  $\lambda$ -weighted net  $\Delta_3^\lambda = \lambda \cdot \Delta_3^{\text{floor}} - \Delta_3^{\text{new}}$ , capturing both the floor reduction from re-individuation and the new within-channel ceiling contribution from the newly-individuated capability. The channels compose into an aggregate alignment-slack reduction bounded above by the sum  $(\Delta_1 + \Delta_2 + \Delta_4) + \Delta_3^\lambda$ , capped at the pre-intervention slack. Each channel formalizes a move the qualitative tradition has been negotiating informally.
6. **Residual-class floor as a shared problem with welfare economics**.  $\varepsilon_{\text{floor}}^{\text{res}}$  is non-zero in any deployment with non-empty ResS, and it is not unique to GFM: the shadow-economy / non-market-activity gap of standard welfare economics [Stiglitz et al., 2009] is the same structural object viewed from a different side. Imputation methods (imputed rent, time-use surveys, satellite accounts) are the existing partial answer; in GFM terms these are Channel 3 re-individuation moves. Alignment-tightening is bounded below by  $\lambda \cdot \varepsilon_{\text{floor}}^{\text{res}}$  regardless of how much within-channel observation infrastructure the deployment invests in.

## 8.2 What this paper defers

- **Per-property Lipschitz constants and the perturbative roadmap.** Theorem 2 is conditional on  $g$  being Lipschitz on the relevant capability subspace with a property-specific  $\text{Lip}(g)$ . The perturbative roadmap of Section 4.4 identifies five candidate alignment properties (anti-monopolar threshold, phase-boundary location, convergence rate, damage bound, aggregate-lower-bound slack) whose Lipschitz analyses would close the per-property bounds. Each requires its own perturbation analysis, which is outside this paper’s scope.
- **Phase-boundary comparison theorem under variable proxy.** Showing the self-correcting basin of [Lasser, 2026f, Phase Redundancy] grows monotonically as  $\varepsilon_{\text{gap}}^{\text{nonres}} \rightarrow 0$  requires a comparison theorem on coupled Lyapunov functions with different reference measures. This is the hardest single piece in the perturbative roadmap and may require new mathematical machinery; Open Question 8.4 states the question precisely.
- **Empirical validation of the structural predictions.** The five predictions of Section 3 and the cross-domain Goodhart test of Section 6.2 are testable in principle but not tested in this paper. The contribution at this stage is to identify the predictions clearly enough that they can be tested and to establish the methodology template through which testing operates.
- **Normative content.** Which capabilities count as central, in Nussbaum’s substantive sense [Nussbaum, 2000, 2011], is outside the formal apparatus. GFM provides the struc-

tural scaffold; the choice of normative loading (which capabilities matter, with which weights, in which contexts) is upstream of the formal apparatus and varies across deployments and societies. The paper takes no position on the normative question.

### 8.3 Structural limitation: Channel 3 can worsen the alignment slack

This is a first-order limitation worth flagging explicitly, and a structural one rather than a deferred-work item. Channel 3 (individuation discipline) contributes to the alignment-slack reduction through the  $\lambda$ -weighted net  $\Delta_3^\lambda = \lambda \cdot \Delta_3^{\text{floor}} - \Delta_3^{\text{new}}$ , which can be *negative* when the newly-individuated capability’s within-channel ceiling contribution exceeds  $\lambda$  times the floor reduction (Remark 29). In that regime, a re-individuation move that looks like “progress” (more of ResS pulled into the observable subset) actually loosens the certified alignment bound rather than tightening it.

This is not a corner case; it is a first-order concern for the empirical programme. Re-individuation is the move welfare economics has been performing for decades through imputation methods, and the current paper does not establish when imputation is alignment-tightening versus alignment-loosening. Concretely: a deployment that introduces a new imputed-time-priced capability for household labour (reducing  $\delta_{\text{residual}}$  by the imputed share) adds the imputation method’s measurement noise to the within-channel ceiling. If the imputation is noisy enough, the deployment’s  $\Delta_3^\lambda$  is negative and the intervention worsens the certified alignment bound. “Tighten all four channels” is therefore not a blanket recommendation; deployments must check the per-channel admissibility, and Channel 3 in particular requires a deployment-specific imputation-quality assessment that the apparatus does not provide. Whether GFM’s machinery (cooperative caps, individuation discipline, downstream cone, polarity boundary) gives sharper imputation tools than welfare economics’ standard methods is the question of Open Question 8.4; the limitation flagged here is that, in the absence of such tools, Channel 3 interventions can backfire, and the paper provides no positive structural guidance for telling when this happens.

### 8.4 Open questions

The following eight questions identify directions for subsequent work and indicate how each would compose with the machinery of this paper.

**OQ 1: The phase-boundary perturbation theorem.** Showing the self-correcting basin of [Lasser, 2026f, Theorem 1] grows monotonically as  $\varepsilon_{\text{gap}}^{\text{nonres}} \rightarrow 0$  requires a comparison theorem on coupled Lyapunov functions with different reference measures. The question is precise: given Lyapunov functions  $V_P$  (defined on the proxy measure  $P$ ) and  $V_T$  (defined on the operational truth  $T$ ), and the basin  $\mathcal{B}_P = \{x : V_P(x) \leq c\}$  for some threshold  $c$ , how does  $\mathcal{B}_T$  relate to  $\mathcal{B}_P$  as a function of  $\varepsilon_{\text{gap}}^{\text{nonres}}$ ? The expected answer:  $\mathcal{B}_T$  differs from  $\mathcal{B}_P$  by a boundary layer of width  $O(\varepsilon_{\text{gap}}^{\text{nonres}})$  with a constant depending on the Lyapunov functions’ smoothness profiles. Establishing the constant is the perturbation-analysis work. This is the hardest single piece of the perturbative roadmap.

**OQ 2: Non-monotonicity risk under multi-dimensional  $\varepsilon_{\text{gap}}^{\text{nonres}}$ .** Definition 3 aggregates relative deviations across capability dimensions via a sup-norm. This is conservative but coarse: a deployment might tighten on some dimensions while loosening on others, and the sup-norm captures only the worst-case dimension. When does multi-dimensional  $\varepsilon_{\text{gap}}^{\text{nonres}}$  admit non-monotonic alignment-tightening? A weighted norm matching the safety-relevance weights  $w_k$  from the alignment property’s Lyapunov function might give a tighter characterization, but the safety-relevance weighting is itself application-dependent. The question is whether a deployment-readable safety-relevance weighting can be constructed for the channels of Section 5, and whether such a weighting changes the channel composition of Proposition 1.

**OQ 3: The Goodhart divergence rate  $f(C)$ .** Conjecture 1 is qualitative: more capable optimizers exploit larger sets of proxy-truth divergence regions, widening the observed  $\varepsilon_{\text{gap}}^{\text{nonres}}$  at equilibrium. A precise functional form  $f : C \mapsto \varepsilon_{\text{gap}}^{\text{nonres}}(C)$  such that  $\varepsilon_{\text{gap}}^{\text{nonres}}(C) \geq f(C) \cdot \varepsilon_{\text{gap}}^{\text{nonres}}(0)$  at equilibrium is optimizer-architecture-dependent. Bounds under specific optimizer models (greedy local search, gradient-based methods, planning-based search, learned-policy optimization) are concrete sub-questions. Each architecture has its own  $f$ , and the structural common content (more reach  $\rightarrow$

more exploitation) is what Conjecture 1 captures without committing to architecture. Closing the architecture- specific bounds is subsequent perturbative-analysis work.

**OQ 4: Completing the Arrow-avoidance comparison.** Section 5.7 argues that GFM *appears* to sidestep Arrow’s original formulation [Arrow, 1951] by being a measure on a poset rather than an ordinal-preference aggregator, with partial analogs of four of the five Arrowian axioms (transitivity, unrestricted domain, Pareto via M3, non-dictatorship via M5) and IIA inapplicable to a volume measure. Sidestepping the original formulation is not the same as proving no analogous impossibility holds. A complete treatment requires (i) a precise formulation of which alternative impossibility results, if any, apply to volume measures on posets, (ii) comparison with Sen’s own ways out [Sen, 1970], and (iii) explicit handling of the heterogeneous-agent aggregation question that Remark 5 flagged as a separate analytical problem. The lineage subsection states the partial argument; the complete version is open.

**OQ 5: Normative loading and alignment-target sensitivity.** GFM provides the structural scaffold; normative choices (“which capabilities matter”) are loaded externally. Two deployments with different normative loadings get different alignment guarantees from the same framework: the economics affects the tightness of the guarantees, but the normative loading affects the target. The question is whether the Lipschitz transfer of Theorem 2 is robust to normative-loading variation, in the sense that the transfer’s content does not depend on the specific loading within a structural admissibility class. A negative answer would mean alignment-tightening claims need to be qualified per-loading; a positive answer would establish that the formalization is normative-loading-invariant within an admissibility class.

**OQ 6: Quantitative rate for  $\varepsilon_{\text{gap}}^{\text{nonres}}$  convergence.** [Lasser, 2026d, Propositions 2 and 3] prove  $\beta^{\text{lower}}$  is non-decreasing under admissible accumulation but do not bound how fast  $\beta^{\text{lower}}$  converges. The rate depends on (i) S0 attestation quality (which itself is institutional-layer- dependent; [Lasser, 2026d, Remark 18]), (ii) Part-A vs Part-B aggregation choice, (iii) observation density on the deployment’s capability inventory, and (iv) the residual-class floor  $\varepsilon_{\text{floor}}^{\text{res}}$  itself. None of (i)–(iv) is bounded in this paper. Bounding the rate is the subsequent perturbative-analysis question for the better-economics-tightens-alignment direction; until it is bounded, Corollary 1 is restricted to direction (monotonicity) rather than rate. This is also [Lasser, 2026e, Open Question 6]’s territory.

**OQ 7: Residual-class floor on tightening, shared with welfare economics.**  $\varepsilon_{\text{floor}}^{\text{res}}$  sets a structural floor below which the within-channel gap component cannot be driven by any amount of observation. This floor is *not unique to GFM*; it is the formal version of the same structural measurement gap that welfare economics has navigated for decades under names like “shadow economy,” “household production,” “non-market activity” [Stiglitz et al., 2009]. Imputation methods (imputed rent, time-use surveys, satellite accounts) are the existing partial answer in welfare economics; in GFM terms, these are Channel 3 re-individuation moves. Characterizing what governs the floor (individuation discipline, governance choices about “no sacrifice-observable path in principle”), and whether GFM’s apparatus (cooperative caps, individuation discipline, downstream cone, polarity boundary) gives sharper imputation tools than the standard methods, is open. This question is structurally distinct from OQ 6: OQ 6 asks how fast the within-channel gap closes; OQ 7 asks how low the floor goes. OQ 7 is shared with welfare economics rather than unique to the alignment programme.

**OQ 8: Adversarial degradation of attestation under optimization pressure.** [Lasser, 2026d, Revealed Sacrifice]’s S0 (calibration), S1 (free choice), and S4(a) (valuation additivity) are not zero-knowledge-verifiable under the default commitment scheme [Lasser, 2026d, Remark 18]; they require institutional attestation. The better-economics-tightens-alignment direction (Corollary 1) depends on attestation *quality*: the framework can drive  $\varepsilon_{\text{gap}}^{\text{nonres}}$  downward only when the attestation layer reliably classifies events under S0, S1, and S4(a). But the same optimization pressure that widens *P-T* divergence regions (Conjecture 1) also incentivizes adversarial degradation of the attestation layer itself: a sufficiently capable optimizer that finds proxy-truth divergence cheaper to exploit than proxy-truth alignment will *also* find attestation manipulation cheaper than honest attestation, given any institutional layer that admits adversarial pressure. Bounding the attestation layer’s adversarial robustness (sybil resistance, collusion resistance, dispute-resolution adequacy under optimizer-controlled inputs) is essential to the operational substance of Corollary 1. This is the

natural follow-up to [Lasser, 2026d, Revealed Sacrifice]’s external-attestation-protocol scope-and-follow-up note, but with the Goodhart-theorem framing: how does attestation quality scale under adversarial pressure, and what incentive-compatibility properties does the layer need to deliver reliable  $\varepsilon_{\text{gap}}^{\text{nonres}}$  reduction in the optimizer-pressure regime? OQ 8 is the natural counterpart on the attestation side to OQ 3 on the optimizer side.

## 8.5 Reading the paper across audiences

The paper is written partly for alignment researchers and partly for welfare economists, with the through-line connecting both audiences: the alignment programme of the GFM sequence is the formalization-payoff of the welfare-economics tradition’s forty-year qualitative negotiation, and progress in either discipline drives the same  $\varepsilon_{\text{gap}}^{\text{nonres}}$  that the other depends on. Some duplication of background is unavoidable, and we have tried to keep technical machinery in clearly delimited sections (the Goodhart theorem and its proof in Section 4.3, the channel composition in Section 5.6) so each audience can navigate to the parts most relevant to their starting context.

For alignment researchers, the load-bearing content is Theorem 2 together with Proposition 1: the proxy-truth gap is structurally controllable through four operationally distinct interventions, each tied to a specific component of the gap decomposition, and each composing into a specific alignment-tightening consequence. The capability-economics lineage of Section 5.7 is the historical-confirmation evidence that the channels are real welfare-economics moves rather than apparatus-internal inventions.

For welfare economists, the load-bearing content is the quantification thesis of Section 5.1 together with the five-features catalogue of Section 2.4: GFM is a measure-theoretic formalization of the Sen–Nussbaum tradition, with axiomatic structure that constrains where Becker’s flexible household-production functions become predictively sharp and that gives the SSF 2009 gap a deployment-readable cell decomposition. The Goodhart theorem is then a result of independent welfare-economics interest: it bounds the cost of optimizing a coarse welfare proxy against a finer operational truth, with the cost decomposing across the same imputation channels welfare economics already uses.

## Author Contributions

**Teague Lasser** owns the paper’s intellectual direction and is responsible for all claims made.

**Claude Opus 4.7** (Anthropic) drafted the paper under that direction.

**GPT 5.5** (OpenAI) served as cold technical reviewer for proof errors and claim mismatches.

*Transparency note.* Both AI systems operated as tools under human direction. Neither system has continuity across sessions, cannot take responsibility for the work in the sense required by most venue authorship policies, and cannot respond to reviewer queries independently. They are listed as authors to accurately represent their contributions to the intellectual content of the paper, not to claim that they meet all criteria of traditional academic authorship. The corresponding author for all inquiries is Teague Lasser.

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## A Dependencies on prior sequence papers

The paper relies on results from nine prior sequence papers ([Lasser, 2026b,g,c,h,a,f,i,d,e]). Table 5 summarises which result is used where.

| Source paper                            | Result imported                                                                                                                                                                                                                           | Role in this paper                                                                                                                                                                                                                                                                                           |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [Lasser, 2026b, Foundation]             | $\text{vol}_P$ as poset measure (Definition 6); naming of B-to-C gap (Section 7)                                                                                                                                                          | The proxy whose proxy-truth gap the paper bounds.                                                                                                                                                                                                                                                            |
| [Lasser, 2026g, Poset Definition]       | Axioms M1–M6 (Proposition 1); $\text{vol}_P$ self-balancing (Theorem 1)                                                                                                                                                                   | The structural apparatus utility theory lacks; M5/M6 carry the non-fungibility, cooperative, individuation features beyond $P_{\text{std}}$ .                                                                                                                                                                |
| [Lasser, 2026c, Horizon Aware]          | Anti-monopolar property (Proposition 6); $\gamma^*$ threshold; exercise pathway (Definition 2); preemptive restriction (Proposition 1)                                                                                                    | Structural prediction with no utility-theoretic analogue; risk machinery the Goodhart theorem perturbs against.                                                                                                                                                                                              |
| [Lasser, 2026h, SCM]                    | Causal-structure formalism                                                                                                                                                                                                                | Substrate for the optimization-pressure argument in Conjecture 1.                                                                                                                                                                                                                                            |
| [Lasser, 2026a, Exogenous Verification] | Wireheading discussion; commitment protocol                                                                                                                                                                                               | Goodhart preview (the original wireheading note); commitment layer composes with [Lasser, 2026d]’s sacrifice channel.                                                                                                                                                                                        |
| [Lasser, 2026f, Phase Redundancy]       | Phase boundary (Theorem 1); design criterion (Theorem 2)                                                                                                                                                                                  | Alignment property whose perturbation under variable proxy is Open Question 8.4.                                                                                                                                                                                                                             |
| [Lasser, 2026i, Controlled Relaxation]  | Convergence (Theorem 2); damage bound (Theorem 1)                                                                                                                                                                                         | Alignment properties listed in the perturbative roadmap (Section 4.4).                                                                                                                                                                                                                                       |
| [Lasser, 2026d, Revealed Sacrifice]     | Aggregate B-to-C lower bound (Theorem 2); monotone accumulation (Propositions 2 and 3); event-local bundle decomposition (Definition 4); residual-class structure; assumptions S0–S5; Remark 18 (external attestation)                    | The measurement channel that drives $\beta^{\text{lower}}$ upward; the formal S0 bridge from inner preference to operational truth; the $\alpha_{n,c}$ substrate for Channel 4; the attestation discipline for Channel 2.                                                                                    |
| [Lasser, 2026e, Need Sufficiency]       | Need-sufficiency architecture (Definition 2); polarity boundary (Definition 3); gap decomposition (Definition 8); wireheading-consistent HHI (Propositions 6 and 7); residual class ResS (Definition 12); need-access cost (Definition 9) | The diagnostic architecture that interprets $\beta^{\text{lower}}$ ; the $\delta$ -decomposition cells through which the four channels of Section 5 act; the polarity boundary that distinguishes need-share from want-share trades; ResS as the formal version of the welfare-economics shadow-economy gap. |

Table 5: Dependencies on prior sequence papers and their roles in this paper. All references are pinned to current preprint formulations; if the upstream preprints revise their numbering or content, downstream references will need refreshing correspondingly.

The capability-welfare-economics references (Sen 1985 [Sen, 1985], Nussbaum 2000 [Nussbaum, 2000], Becker 1965 [Becker, 1965], Stiglitz–Sen–Fitoussi 2009 [Stiglitz et al., 2009], Arrow 1951 [Arrow, 1951]) are positioning rather than load-bearing in the formal sense: their qualitative content is what the GFM apparatus formalizes, and the formalization-claim does not technically depend on any specific result from these references. Section 5.7 makes the structural correspondences explicit.